Model 2.0

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Part I. Model 2.0

1 How Representations Shape Strategic Behavior

Two players interact once. Player S is the first mover—the dictator, the proposer, the investor; generically, the *sender*—and player R is the second mover or *receiver*. The sender's initial wealth is normalized to one, and her action is the share $t \in [0,1]$ she sends to the receiver. Total final wealth is $W(t)$, which may depend on the amount sent, π_S denotes the sender's material payoff, and π_R denotes the receiver's material payoff.

A *representation* of the game is a pair $\mathbf{r} = (\boldsymbol{\alpha}, \mathbf{x})$: an attention profile $\boldsymbol{\alpha} = (\rho, \sigma, \mu)$ over payoff elements and, in the strategic games, a believed behavior $\mathbf{x}$ of the receiver. Given her representation, the sender maximizes

$$u_S(t) \;=\; \rho\, W(t) \;-\; \sigma\, \pi_R \;-\; \frac{\mu}{2}\,(t - t^e)^2\,, \tag{1}$$

where the three payoff elements are total wealth, the receiver's material payoff, and a sharing norm anchored at the *equal-payoff action* t^e—the action that equalizes the two players' material payoffs under the sender's representation: the equal split $t^e = \frac{1}{2}$ in the dictator and ultimatum games and, in the trust game, where the multiplier breaks that coincidence, the send that equalizes expected payoffs at the believed return (Proposition 3). The Greek letters are attention weights, with $\rho = \sigma = \mu = 1$ the undistorted case in which no motive is slanted by the representation and the sender has inequity-averse utility in the spirit of Bolton and Ockenfels (2000). Standard models prescribe *stability*—that is, an attention profile that stays fixed while context varies—and, in the strategic games, *rational expectations*—that is, beliefs that track actual behavior.

Because $\pi_S = W(t) - \pi_R$, equation (1) can be rewritten as

$$u_S(t) \;=\; \sigma\, \pi_S \;+\; (\rho - \sigma)\, W(t) \;-\; \frac{\mu}{2}\,(t - t^e)^2\,, \tag{2}$$

which makes the mapping to motives transparent: σ is attention to own earnings, $\rho - \sigma$ is attention to the total surplus over and above one's own share, and μ is attention to the norm. The functional form thus nests the motives of standard distributional preference models (Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000; Charness and Rabin, 2002), with two differences. First, the moral term is *idealistic*: it attaches to what the sender does, not to what materializes afterwards. Whether the receiver accepts, rejects, or reciprocates, the sender has still sent the same t; only material payoffs are contingent on the receiver's move. This assumption embodies a simple moral psychology—one feels right or wrong about one's own conduct—and makes the strategic games tractable. Second, beliefs are part of the representation rather than equilibrium objects: the process that determines what the game is about also determines whom one imagines playing against. The attention weights and beliefs are properties of the situation the game evokes, not stable traits: the same person can be a moral dictator and a self-interested price-setter. Proofs of all propositions are in Appendix A.

Dictator Game. Here $W(t) = 1$ and $\pi_R = t$; there is no second mover to forecast, so $\mathbf{x}$ is empty.

Proposition 1 (Dictator Transfer)**.** *The optimal transfer is* $t^{DG} = \max\left\{0,\ \frac{1}{2} - \frac{\sigma}{\mu}\right\}$*. It never exceeds one half, is decreasing in the selfishness ratio* σ/μ*, and is independent of* ρ*.*

A moral representation (high μ) anchors the transfer at the equal split; a self-interested one (high σ, low μ) drives it to zero. A cooperative representation pins down ρ, which is idle in a fixed-pie game with no strategic margin: cooperative dictators act on their residual σ/μ.

Ultimatum Game. If R accepts, payoffs are as in the dictator game; if R rejects, both players earn zero. The sender's representation now includes a believed behavior of the receiver, which we embed in an *acceptance schedule*: the sender believes an offer t is accepted with probability

$$p(t) \;=\; a + b\,t, \qquad a, b > 0,$$

so the belief component of her representation is the pair $\mathbf{x} = (a, b)$. Because the moral payoff is idealistic, only the material part of utility is scaled by acceptance, and the sender solves

$$\max_t \;\; p(t)\,\big(\rho - \sigma t\big) \;-\; \frac{\mu}{2}\,\big(t - \tfrac{1}{2}\big)^2. \tag{3}$$

Proposition 2 (Ultimatum Offer)**.** *The objective in* (3) *is strictly concave, and the optimal offer is*

$$t^{UG} \;=\; \frac{1}{2} \;+\; \frac{b\rho - (a+b)\,\sigma}{2b\sigma + \mu}, \tag{4}$$

censored to $[0, 1]$*; the offer is interior if and only if* $\left|b\rho - (a+b)\,\sigma\right| < b\sigma + \mu/2$*. At an interior optimum: (i) if* $p(1) = a + b \leq 1$*, then* $t^{UG} > t^{DG}$ *for every attention profile; (ii) the offer is increasing in* ρ*, decreasing in* σ*, and moved toward the equal split by* μ *(raised when* $t^{UG} < \frac{1}{2}$*, lowered when* $t^{UG} > \frac{1}{2}$*); (iii) the offer is decreasing in believed acceptance* a*, with sensitivity* $\left|\partial t^{UG}/\partial a\right| = \sigma/(2b\sigma + \mu)$*, which is increasing in* σ *and decreasing in* μ*.*

Part (i) is the strategic discipline of the ultimatum game: because low offers risk rejection, *every* representation offers more than it would transfer as dictator. Part (iii) delivers two sharp empirical predictions. First, holding the representation fixed, offers should *fall* with believed acceptance—a proposer who thinks low offers will be accepted makes them. Second, the belief sensitivity itself depends on attention: a moral (high-μ) proposer acts on the norm regardless of what she expects, while a self-interested (high-σ) proposer tracks her beliefs. Attention to total wealth also acquires a role it lacks in the dictator game: $\partial t^{UG}/\partial \rho = b/(2b\sigma + \mu)$, which is strictly positive yet vanishes when acceptance is insensitive to the offer ($b = 0$). The pie is fixed, but rejection destroys it, so weight on total wealth raises the offer purely through the fear of that destruction.

Remark 1 (Endogenous Acceptance Schedule). The linear acceptance schedule can be founded in a moral psychology similar to the one we introduced above for senders. Endow the receiver with attention weights $\hat{\boldsymbol{\alpha}} = (\hat{\rho}, \hat{\sigma}, \hat{\mu})$. Assume that the idealistic principle is applied to the sender's action: the receiver suffers from the offer's inequity and feels this cost whether or not the offer is accepted. Moreover, assume that rejection delivers a psychological benefit of *protest*, $\psi \geq 0$, which is drawn from a distribution F and weighted by moral attention. In this case, the inequity-aversion term cancels from the accept–reject margin and the receiver accepts t if and only if $\hat{\rho}-\hat{\sigma}(1-t) \geq \hat{\mu}\,\psi$, so the acceptance probability is $F\big[(\hat{\rho}-\hat{\sigma}+\hat{\sigma}t)/\hat{\mu}\big]$. For ψ uniform on $[0,\bar{\Psi}]$ this is exactly $p(t)=a+bt$ with

$$a=\frac{\hat{\rho}-\hat{\sigma}}{\hat{\mu}\bar{\Psi}},\qquad b=\frac{\hat{\sigma}}{\hat{\mu}\bar{\Psi}}.$$

The believed acceptance schedule (a,b) is therefore a representation of the *receiver's motives*: imagining a punitive receiver (high $\hat{\mu}\bar{\Psi}$—the *moral bad* category) scales acceptance down across the board, while imagining a self-interested receiver (high $\hat{\sigma}$) makes acceptance sensitive to money. Note also that $p(1)=a+b=\hat{\rho}/(\hat{\mu}\bar{\Psi})$, so the condition $a+b\leq 1$ in Proposition 2 has a natural reading: the imagined receiver's protest weight is large enough that even a full-split offer is not certain to be accepted.

Trust Game. The amount sent is multiplied by $1+r$, with $r>0$ the net return, so total wealth is $W(t)=1+rt$. The sender believes the receiver returns a share s, with $0\leq s<1$, of the multiplied amount $(1+r)t$, so the belief component of her representation is $\mathbf{x}=s$ and the receiver's believed material payoff is $\pi_R=(1-s)(1+r)t$. We assume that s is constant; Remark 2 below founds the return schedule in the receiver's moral psychology, and Proposition 4 shows that the comparative statics survive when the believed share declines in the send. Expected payoffs equalize at the send $t^e(s)=\big[1+(1+r)(1-2s)\big]^{-1}$, which is increasing in the believed return and equals one half exactly at $s=r/\big(2(1+r)\big)$—one third of the output at $r=2$—so here the equal-payoff anchor moves with the belief. The sender solves

$$\max_t\ \ \rho\,(1+rt)\ -\ \sigma\,(1+r)\,t\,(1-s)\ -\ \frac{\mu}{2}\,(t-t^e(s))^2\,. \tag{5}$$

Proposition 3 (Trust Send)**.** *The optimal send is*

$$t^{TG}\ =\ t^e(s)\ +\ \frac{\rho r-\sigma(1+r)(1-s)}{\mu}, \tag{6}$$

censored to $[0,1]$*; the send is interior if and only if* $-\mu\,t^e(s)<\rho r-\sigma(1+r)(1-s)<\mu\big(1-t^e(s)\big)$*. At an interior optimum, the send is increasing in* ρ*, decreasing in* σ*, and moved toward the equal-payoff send* $t^e(s)$ *by* μ*; it is increasing in the believed return share* s *with sensitivity* $\mathrm{d}t^e/\mathrm{d}s+\sigma(1+r)/\mu$*, where* $\mathrm{d}t^e/\mathrm{d}s=2(1+r)\big[1+(1+r)(1-2s)\big]^{-2}>0$*: beliefs move the send through the anchor itself as well as through the material term, so the sensitivity is strictly positive even under norm domination. As* $\mu\to 0$ *the problem becomes linear and the solution goes to a corner: a purely material sender sends everything if she believes the returned share exceeds the break-even level,* $s(1+r)>1$*, and nothing if it falls short; the anchor is likewise censored—*$t^e(s)\geq 1$ *for* $s\geq\frac{1}{2}$*—so a norm-dominated sender who expects at least half of the output back is pulled to the full send. Finally, at interior optima,* $t^{TG}\geq t^{DG}$ *if and only if* $r\big[\rho-\sigma(1-s)\big]+\sigma s\geq\mu\big(\frac{1}{2}-t^e(s)\big)$*; for* $s\geq r/\big(2(1+r)\big)$ *the right side is non-positive and* $\rho\geq\sigma(1-s)$ *suffices.*

The trust game is where both components of the representation bite at once. Attention to total wealth is no longer idle: ρ multiplies the return r, so cooperative senders—those who attend to the multiplied pie itself—send the most. Moral senders are anchored near the equal-payoff send. Self-interested senders send the least unless they are optimistic, and their sends are the most belief-sensitive—though the anchor itself moves with the belief, so even norm-dominated senders track the expected return through $\mathrm{d}t^e/\mathrm{d}s$. Whether the send sits above or below the anchor—the sign of the second term in (6)—is ambiguous: it is positive if and only if $\rho/\sigma>(1+r)(1-s)/r$, and for a purely material sender ($\rho=\sigma$) this reduces to the break-even condition $s(1+r)>1$, so the term is positive exactly when she believes trusting pays. The belief sensitivity $\mathrm{d}t^e/\mathrm{d}s + \sigma(1+r)/\mu$ holds at any interior optimum regardless of this sign (at a censored send it is locally zero). And in contrast to the ultimatum game, where beliefs enter with a *negative* sign, beliefs now enter with a *positive* sign: sender's optimism about the return in the trust game raises the optimal send, while sender's optimism about acceptance in the ultimatum game lowers the optimal offer.

Remark 2 (Endogenous Return Schedule). The believed return can be founded in the same moral psychology as the acceptance schedule of Remark 1. Endow the imagined receiver with attention weights $\hat{\boldsymbol{\alpha}} = (\hat\rho, \hat\sigma, \hat\mu)$. Holding the output $(1+r)t$, the receiver is a dictator over it: total wealth $1+rt$ is fixed no matter how much she returns, so $\hat\rho$ is idle and she trades own earnings against a norm of returning half of a target pie X, choosing the returned amount y to maximize $\hat\rho(1+rt) - \hat\sigma(1-t+y) - \frac{\hat\mu}{2}\big(y/X - \frac{1}{2}\big)^2$. The natural targets are the amount sent ($X = t$), the output ($X = (1+r)t$), and total wealth ($X = 1+rt$), and for every target the optimal returned share of the target is the same expression, $y^*/X = \frac{1}{2} - (\hat\sigma/\hat\mu)X$. Under the first two targets the believed returned share of the output is affine and *decreasing* in the send—with the amount-sent target, $s(t) = \big[\frac{1}{2} - (\hat\sigma/\hat\mu)t\big]/(1+r)$; with the output target, $s(t) = \frac{1}{2} - (\hat\sigma/\hat\mu)(1+r)t$. Thus, the foundation has a testable signature: believed return shares should decline in the send, which the two elicited belief points per sender (reference and chosen action) can check. The total-wealth target is ill behaved—at small sends the norm target exceeds the pot itself, forcing full return below a threshold and a kinked schedule—so the amount-sent and output targets are the tractable specifications. As in Remark 1, the believed schedule is a representation of the receiver's motives, here through the imagined selfishness ratio $\hat\sigma/\hat\mu$.

Replacing the constant share with the founded, declining schedule—or any affine one—leaves the trust-game analysis intact:

Proposition 4 (Trust Send with a Declining Return Schedule)**.** *Under any affine believed schedule $s(t) = s_0 - s_1 t$ with $0 \le s_0 < 1$ and $s_1 \ge 0$, expected payoffs equalize at the unique positive root t^e of $2(1+r)s_1x^2 + \big[1 + (1+r)(1-2s_0)\big]x - 1 = 0$, which reduces to $t^e(s_0)$ at $s_1 = 0$. The sender's objective remains strictly concave, and the optimal send is*

$$t^{TG} \;=\; \frac{\mu\, t^e + \rho r - \sigma(1+r)(1-s_0)}{\mu + 2\sigma(1+r)s_1}, \tag{7}$$

censored to $[0,1]$, which nests equation (6) at $s_1 = 0$; the send is interior if and only if $-\mu\, t^e < \rho r - \sigma(1+r)(1-s_0) < \mu(1-t^e) + 2\sigma(1+r)s_1$. At an interior optimum: (i) the send is increasing in ρ, decreasing in σ, and moved toward the equal-payoff send t^e by μ, as in Proposition 3; (ii) the send is increasing in the believed level s_0, with sensitivity $\big[\mu\,\partial t^e/\partial s_0 + \sigma(1+r)\big]/\big(\mu + 2\sigma(1+r)s_1\big)$—damped by the believed slope exactly as the ultimatum sensitivity $\sigma/(2b\sigma + \mu)$ is damped by b, with constant s the undamped limit; (iii) the send is decreasing in the believed slope, $\partial t^{TG}/\partial s_1 = \big[\mu\,\partial t^e/\partial s_1 - 2\sigma(1+r)\,t^{TG}\big]/\big(\mu + 2\sigma(1+r)s_1\big) < 0$: greater imagined receiver selfishness lowers both the anchor and the material pull of sending.

2 Summary of Comparative Statics

Table 1 collects the comparative statics.

Table 1: **Summary of Comparative Statics**

	Attention and Beliefs				Belief Sensitivity
Transfer	ρ	σ	μ	Belief	$\lvert \partial t/\partial \text{Belief} \rvert$
Dictator t^{DG}	0	$-$	$\to \frac{1}{2}$	NA	NA
Ultimatum t^{UG}	$+$	$-$	$\to \frac{1}{2}$	$-$	$\sigma/(2b\sigma+\mu)$
Trust t^{TG}	$+$	$-$	$\to t^e(s)$	$+$	$\mathrm{d}t^e/\mathrm{d}s + \sigma(1+r)/\mu$

Notes: Signs at interior optima. Solutions are censored to $[0,1]$. "$\to \frac{1}{2}$" ("$\to t^e(s)$") means the moral weight moves the transfer toward the norm anchor—the equal split in the fixed-pie games, the equal-payoff send in the trust game—so over the empirically relevant range its effect is positive. In the ultimatum game, beliefs are about the constant of the acceptance schedule, a. In the trust game, beliefs are about the constant return share, s. In both strategic games, the material channel of the belief sensitivity is increasing in σ and decreasing in μ; in the trust game the anchor itself moves with the believed return, adding the weight-free term $\mathrm{d}t^e/\mathrm{d}s$, so even norm-dominated senders track beliefs there.

3 Welfare and Inequality

The comparative statics about the sender's action immediately deliver comparative statics for total surplus and for the division of payoffs, which is where representations acquire social content. Let V_g denote expected total material wealth at an interior optimum—$V_{DG} = 1$, $V_{UG} = p(t^{UG})$ (the pie survives only if the offer is accepted), $V_{TG} = 1 + r\,t^{TG}$—and let $D_g = \pi_S - \pi_R$ denote the payoff gap conditional on trade (in the ultimatum game, given acceptance; the other games have no rejection margin).

Proposition 5 (Welfare)**.** *At interior optima: (i) surplus attention raises welfare where the pie responds to the action, with strength quadratic in that response: $\partial V_{DG}/\partial\rho = 0$, $\partial V_{UG}/\partial\rho = b^2/(2b\sigma+\mu)$, $\partial V_{TG}/\partial\rho = r^2/\mu$; (ii) own-payoff attention destroys welfare in both strategic games, $\partial V_{UG}/\partial\sigma = -b\big[(a+b)\mu + 2b^2\rho\big]/(2b\sigma+\mu)^2 < 0$ and $\partial V_{TG}/\partial\sigma = -r(1+r)(1-s)/\mu < 0$, and unambiguously lowers the counterpart's expected payoff; the sender's own expected payoff can fall as well, precisely when $b(1-2t^{UG}) > a$—a steep believed schedule with a low intercept; (iii) in the strategic games, norm attention raises welfare exactly when the material pull is selfish—$\partial V_g/\partial\mu$ has the sign of $t^e - t^g$, with $t^e = \frac{1}{2}$ in the fixed-pie games—while dictator-game welfare is unaffected by any weight; (iv) optimism raises welfare: $\partial V_{UG}/\partial a = (b\sigma+\mu)/(2b\sigma+\mu) > 0$ and $\partial V_{TG}/\partial s = r\big[\mathrm{d}t^e/\mathrm{d}s + \sigma(1+r)/\mu\big] > 0$.*

Part (i) formalizes cooperation through attention: a sender who attends to total wealth gives more precisely where giving protects (ultimatum) or produces (trust) the pie, and welfare rises with the *square* of the pie's sensitivity to the action, because attention moves the action by that sensitivity and the action moves the pie by it again. Part (ii) is the reverse face: own-payoff attention is socially destructive in every strategic game, and it is not even reliably self-serving—when rejection risk is steep, the offer cut costs more in expected acceptance than it saves in shares. Part (iii) says the norm is a social technology against selfish material pulls, not a complement to cooperative ones: when the pull is generous (an optimistic cooperator in the trust game), the anchor drags the send back toward the equal-payoff point and destroys surplus—though the drag is milder for optimists, whose anchor itself sits above one half.

These statements evaluate the ultimatum pie at the sender's own beliefs: $V_{UG} = p(t^{UG})$ is expected surplus under her model of the world. This is by design—a rational-expectations welfare measure would impose exactly the belief–behavior alignment the paper denies—and the ultimatum game is the only place where the choice matters: dictator surplus is constant, and trust surplus $1 + r\, t^{TG}$ is belief-free given the send, with the division evaluated at the actual share below.

Remark 3 (Believed versus Realized Welfare). Let actual acceptance follow any increasing, differentiable schedule $\hat{p}$, held fixed as the sender's parameters vary, and let $\hat{V}_{UG} = \hat{p}(t^{UG})$ denote realized expected surplus at the offer chosen under the believed schedule. Parts (i)–(iii) of Proposition 5 survive with the same signs: attention weights enter neither schedule, so they move realized surplus only through the action, $\partial \hat{V}_{UG}/\partial\theta = \hat{p}'(t^{UG}) \cdot \partial t^{UG}/\partial\theta$ for $\theta \in \{\rho, \sigma, \mu\}$, and only magnitudes change. Part (iv) reverses in the ultimatum game: optimism no longer raises acceptance directly—it only lowers the offer, and actual acceptance falls with it—so $\partial \hat{V}_{UG}/\partial a < 0$; its trust-game half is unchanged.

Believed and realized welfare thus part ways exactly where beliefs and behavior do—an optimistic proposer expects more surplus and produces less—which is the forecast-error mechanism behind the ultimatum-game surplus accounting in the paper.

Proposition 6 (Inequality)**.** *Conditional on trade, $D_{DG} = 2\sigma/\mu$ and $D_{UG} = 2\big[(a + b)\sigma - b\rho\big]/(2b\sigma + \mu)$: own-payoff attention widens the gap, surplus attention narrows it (strictly in the ultimatum game), and norm attention compresses it toward zero, $\partial D_{UG}/\partial\mu = -D_{UG}/(2b\sigma + \mu)$ and $\partial D_{DG}/\partial\mu = -D_{DG}/\mu$. In the trust game, for an actual returned share s_a, $D_{TG} = 1 - t\big[1 + (1 + r)(1 - 2s_a)\big]$: a larger send narrows the gap if and only if $s_a < (2 + r)/\big(2(1 + r)\big)$, and norm attention compresses the gap toward its value at the anchor, $1 - t^e(s)\big[1 + (1 + r)(1 - 2s_a)\big]$, which is zero precisely when the belief is correct ($s = s_a$): the norm eliminates conditional inequality exactly when the sender forecasts the return correctly, and the residual gap at the anchor measures her belief error.*

The equal-payoff anchor coincides with the equal split in the fixed-pie games but not in the trust game: with $r = 2$—the experiment's tripling of the send—the equal-split send equalizes payoffs only if the receiver returns exactly one third of her output, and the anchor $t^e(s)$ instead moves with the believed return. The trust game is thus where a payoff-based sharing norm separates from an action-based one. The population reading connects these statics to the treatments: a treatment that reweights retrieved categories toward high-σ, low-μ representations in the ultimatum game destroys expected surplus and widens conditional inequality, while one that reweights toward high-ρ representations and optimistic beliefs in the trust game creates surplus—the pattern of the surplus accounting in the paper.

4 From Measured Categories to Model Parameters

The empirical categories map onto the representation as follows:

- A **Moral** representation corresponds to high norm attention μ.
- A **Self-Interest** representation corresponds to high own-earnings attention σ with $\rho \approx \sigma$. Indeed, when the surplus weight $\rho - \sigma$ is nil, own earnings are $\sigma\pi_S$.
- A **Mutual Benefit/Cooperation** representation corresponds to surplus attention, $\rho > \sigma$.

For second movers, Remark 1 distinguishes **Moral good** (fairness, reciprocity: $\hat{\rho} > \hat{\sigma}$ with moderate protest weight $\hat{\mu}\bar{\Psi}$) from **Moral bad** (a high protest weight $\hat{\mu}\bar{\Psi}$, hence rejection), with **Self-Interest** making acceptance money-sensitive and returns low. The belief components (a, b)

and s are measured directly by the belief and hypothetical-belief questions. The model does not restrict how $\mathbf{x}$ relates to $\boldsymbol{\alpha}$ across people: beliefs may be anchored on a commonly held standard or may covary with own attention.

These mappings can be taken to the data. The categories themselves remain measured from text; choices enter only to quantify the attention profile each category carries. Behavior identifies attention only up to scale—a common rescaling of (ρ, σ, μ) leaves choices unchanged—so the objects to calibrate are the ratios $(\rho/\mu, \sigma/\mu)$, category by category, from control moments. The dictator transfer pins down selfishness: $\sigma/\mu = \frac{1}{2} - t^{DG}$ for interior transfers. The trust send and the measured mean believed share $\bar{s}$ then pin down surplus attention: $\rho/\mu = \left[t^{TG} - \frac{1}{2} + (\sigma/\mu)(1+r)(1-\bar{s})\right]/r$. The ultimatum game overidentifies: given $(\rho/\mu, \sigma/\mu)$, the offer's first-order condition together with the measured reference-action acceptance belief implies an acceptance schedule (a, b), which must be a genuine probability schedule ($b > 0$, $a + b \leq 1$) and—because measured acceptance beliefs are flat across categories—should come out similar across categories; the implied belief sensitivities $\sigma/(2b\sigma + \mu)$ and $\sigma(1+r)/\mu$ can then be compared with the estimated category-by-belief interaction slopes. Two limits of the exercise are themselves predictions: for a norm-dominated category the offer sits at the anchor and the inversion carries almost no information about (a, b)—beliefs are unidentified exactly where the model says behavior is insensitive to them—and a category rare in the dictator game (Mutual Benefit/Cooperation) leaves a one-dimensional locus rather than a point, though the trust moment alone already locates that locus in the region $\rho > \sigma$ for every admissible σ/μ.

Results (2026-07-19 run, equal-payoff anchor; replication package, `08_calibration.py`). Self-interest: $\sigma/\mu = 0.397$, $\rho/\mu = 0.426$—so $\rho \approx \sigma$ *emerges*, it is not imposed—with FOC-implied schedule $(a, b) = (0.324, 0.481)$, a genuine probability schedule ($a + b = 0.81 \leq 1$). Moral: $\sigma/\mu = 0.038$, $\rho/\mu = 0.022$; the inversion returns $b < 0$—no valid schedule, i.e., beliefs unidentified exactly as norm-domination implies. Mutual Benefit/Cooperation (UG and TG conditions solved jointly, imposing the common schedule licensed by flat reference-action beliefs): $\sigma/\mu \approx 0$, $\rho/\mu = 0.048$, on the $\rho > \sigma$ locus throughout. Sensitivities: in the ultimatum game, predicted 0.29 for Self-interest against estimated 0.11 (attenuation ≈ 0.4); in the trust game, where the anchor's belief channel $\mathrm{d}t^e/\mathrm{d}s$ is common to all types, predicted 1.46 (Moral), 1.74 (MBC), and 2.03 (Self-interest) against estimated 0.31, 0.26, and 0.45—attenuation uniform at 0.15–0.22 across all three cells, with no cell deviating in the other direction. Bootstrap SEs and the full log are in `calibration_stats.txt`.

The two-point route fails, informatively. The two elicited belief points per proposer—hypothetical acceptance at the reference action and the belief at the chosen action—could in principle identify (a, b) from measurement alone (category-level means, when the mean chosen offer differs from the reference action), freeing the first-order condition to serve as a genuine overidentifying test and identifying the schedule even for norm-dominated categories. In the data this route *fails validity*: it implies $a < 0$ and $a + b$ between 1.7 and 2.6 for Moral and Self-interest, because beliefs at chosen offers are 16–20 percentage points above any coherent linear schedule through the reference-action point. Read as measurement, the failure says that chosen-action beliefs embed optimism or selection at the chosen action; the headline calibration therefore uses the FOC-inversion route and keeps the chosen-offer belief as the overidentifying test, whose uniform rejection—measured beliefs exceed the schedule at chosen offers in every category—is itself content for the forecast-error analysis.

Remark 4 (Joint Distribution of Attention and Beliefs). Because attention and beliefs are both prototype content, the mixture over categories generates a *joint* distribution of $(\boldsymbol{\alpha}, \mathbf{x})$, and the games aggregate that joint in a known way—so the attention–belief coupling is itself testable content, not a free object. Compare mean behavior in a heterogeneous population with the counterfactual in which attention profiles and beliefs are drawn independently from their marginals. At interior optima the trust send is bilinear in the selfishness ratio and the believed share, so the entire dependence of the mean send on the coupling runs through a single moment:

$$E\big[t^{TG}\big] - E_{\perp}\big[t^{TG}\big] \;=\; (1+r)\,\mathrm{Cov}\!\left(\frac{\sigma}{\mu},\, s\right),$$

where $E_{\perp}$ holds both marginals fixed (ρ/μ enters the send linearly, so no other covariance moves the mean). Beliefs anchored on a commonly held standard imply a zero covariance—the marginals suffice—while projection, a self-interested sender imagining an ungenerous receiver, implies a negative covariance: mean sends fall short of the independence counterfactual. No such formula exists for the ultimatum game, where the believed slope enters the denominator $2b\sigma+\mu$ and the independence gap must be computed numerically—though with acceptance beliefs common across categories the model predicts it is close to zero; likewise, under the declining schedule of Remark 2 exactness obtains only in the constant-s limit $s_1 = 0$. Category-level calibration supplies the joint directly—each category carries an attention profile and a belief—so the covariance and the counterfactual are byproducts of the exercise just described. In the data (control trust game; category-level σ/μ, individual believed shares), $\mathrm{Cov}(\sigma/\mu, s) = -0.008$: mean sends sit 2.3 points of the endowment below the independence counterfactual—the projection sign, with the between-category version delivering the same number (2026-07-19 run, `08_calibration.py`).

5 How Representations Are Determined

The sender does not see a game form; she sees a *description* $\mathbf{d} = (\mathbf{r}_d, \boldsymbol{\kappa}_d)$, where $\mathbf{r}_d$ is the representation implied by a literal reading of the rules and $\boldsymbol{\kappa}_d$ is the context: the labels, the cover story, the setting in which the interaction is embedded. Her memory contains *categories* of experience $c \in \mathbf{C}$—classes of situations she has lived or can imagine, such as sharing a windfall, haggling over a price, investing in a joint project—each with a typical representation $\mathbf{r}_c$, a typical context $\boldsymbol{\kappa}_c$, and a time-weighted frequency F_c (Bordalo et al., 2020, 2026). Facing the description, she selects a category and a representation $\mathbf{r} = (\boldsymbol{\alpha}, \mathbf{x})$ to solve

$$\max_{c\in\mathbf{C},\,\mathbf{r}\in\mathbf{R}} \;\; F_c \cdot S\big[(\mathbf{r}, \boldsymbol{\kappa}_d), (\mathbf{r}_c, \boldsymbol{\kappa}_c)\big] \;+\; S\big[(\mathbf{r}, \boldsymbol{\kappa}_d), (\mathbf{r}_d, \boldsymbol{\kappa}_d)\big] \;+\; \epsilon_c, \tag{8}$$

where S is a similarity function (Tversky, 1977; Gilboa and Schmeidler, 1995) and ϵ_c is an i.i.d. extreme-value shock. The chosen representation trades off fidelity to the description (the second term) against assimilation to the best-fitting category of experience (the first term).

Equation (8) has two forces, and each of the experimental manipulations targets one of them. First, a higher frequency F_c fosters a category's retrieval, increasing the likelihood that the sender slants her representation toward $\mathbf{r}_c$, *for a given description*. This is the **story lever**: reading and summarizing a story about a workplace bonus or about charitable aid is a recent, vivid experience that raises the retrieval weight of its category, while the description of the game itself is unchanged. Second, a change in the context $\boldsymbol{\kappa}_d$ that makes the description more similar to the contexts $\boldsymbol{\kappa}_c$ of experiences in some category also fosters that category's retrieval, *for given frequencies*. This is the **frame lever**: describing the identical game as a transaction between a data provider and a broker moves the description itself into the neighborhood of market experiences.

The frequency weight has a natural recency structure. Let each experience e in category c have age τ_e and let $F_c = \sum_{e\in c} \delta^{\tau_e}$ with $\delta \in (0, 1)$: frequency is experience discounted by recency. The description just read is then itself the most recent trace—age zero, weight $\delta^0 = 1$—which is why fidelity carries a fixed unit weight in equation (8): both terms are recency-weighted similarities. This reading disciplines the trade-off between memory and description. A story is an age-zero experience added to its category, so a single story can rival a lifetime of discounted experience—which is why the story lever works at all; and when discounting is strong every F_c is small, so the description dominates and memory cannot drown it unless a category is both frequent and recent. A natural refinement scales fidelity by a cue-strength parameter increasing in the description's distinctiveness—its dissimilarity from the stored contexts—so that a description unlike anything in memory resists assimilation.

Three implications organize the empirical analysis.

1. *Heterogeneity within a context.* Within a treatment, people hold different representations—they have different histories F_c and draw different shocks ϵ_c—so behavior is heterogeneous, and its heterogeneity is structured by the mixture of retrieved categories, not merely by noise.

2. *Instability across contexts.* Across treatments, the distribution of retrieved categories shifts—through F_c for the stories, through $\boldsymbol{\kappa}_d$ for the frames—and the action moves in the direction that combines the attention differences (across $\boldsymbol{\alpha}$) and the belief differences (across $\mathbf{x}$) of the categories gaining and losing weight. Holding category-typical actions fixed, the mean effect of a manipulation is the composition change $\Delta E[t^g] = \sum_c \Delta q_c \, \bar{t}_c^g$, where q_c is the retrieval probability of category c and $\bar{t}_c^g$ its typical action in game g: the product of a retrieval shift, whose size is gated by the similarity between the game's description and the manipulated content, and the spread of category-typical actions in that game. Both factors are game-specific and measurable, so the same manipulation can move one game strongly and another barely at all—not because any payoff element drops out of the representation, but because retrieval shifts less where similarity is low and matters less where the retrieved categories act alike.

3. *Forecast errors and learning.* Nothing in (8) guarantees that the two players' representations are shared or mutually consistent: the sender's imagined receiver is retrieved from the sender's memory, not from the receiver's behavior. Systematic, context-dependent forecast errors are therefore an equilibrium object of the theory, not a pathology—and learning in games under a stable context can be read as convergence toward representations that no longer generate salient surprises, which is what renders equilibria stable.

The theory's inputs are, in principle, measurable: the similarity of the experimental contexts to broad categories of experience—cooperation problems, conflict problems, moral problems—and to receiver types who accept or reject, return or keep.

6 Receiver's Behavior

So far the receiver exists only as the sender's belief. The same two building blocks—moral psychology given a representation (Remarks 1 and 2) and retrieval given a description (equation (8))—turn her into a full player. The receiver's description differs from the sender's in one crucial way: it contains the sender's action. She faces $\mathbf{d}_R = (\mathbf{r}_d, \boldsymbol{\kappa}_d, t)$—not an abstract game, but a game in which she has just been offered t or entrusted $(1+r)t$. Retrieval then assigns category c a probability $q_c(t)$ (a logit, under the extreme-value shocks), *with the sender's action as a contextual cue*, and conditional on the category, behavior follows the second-mover margins already derived—reading the hatted weights of Remarks 1 and 2 as the receiver's own retrieved attention profile: an acceptance schedule

$p_c(t) = a_c + b_c t$ from the protest psychology, and a returned share from the dictator-over-the-output problem. For the return margin the equal-payoff anchor matters: the receiver moves last, so her norm anchors at the *realized* equalizing return—the share $s^e(t) = \frac{2+r}{2(1+r)} - \frac{1}{2(1+r)t}$ of the output, the inverse of the sender's anchor ($t^e(s^e(t)) = t$), rising in the send and reaching zero at $t = 1/(2+r)$—and trading the anchor against own earnings gives $s_c(t) = \max\{0,\ s^e(t) - (\hat{\sigma}/\hat{\mu})_c(1+r)t\}$: near-equalizing and *rising* schedules for low-selfishness categories, low and flat ones as $(\hat{\sigma}/\hat{\mu})_c$ grows. (Anchoring instead at half of a target pie, as in the belief foundation of Remark 2, gives affine *declining* schedules—the two anchors are separable in the data.) Aggregate second-mover behavior is the mixture

$$p(t) = \sum_c q_c(t)\, p_c(t), \qquad s(t) = \sum_c q_c(t)\, s_c(t). \tag{9}$$

The sender's action now works through two channels,

$$p'(t) \;=\; \underbrace{\sum_c q_c(t)\, p'_c(t)}_{\text{margin within categories}} \;+\; \underbrace{\sum_c q'_c(t)\, p_c(t)}_{\text{composition across categories}},$$

and if low offers retrieve condemnation—an insulting offer resembles situations of exploitation, so $q_{\text{Moral bad}}$ falls with t and the released probability mass flows to categories with higher acceptance—the composition term is positive and the aggregate schedule is steeper than the within-category margins alone, and can be steeper than *every* category schedule. Three consequences follow. First, measured acceptance can respond to offers more strongly than any fixed protest psychology implies: part of the response is a change in what the offer makes the receiver think the situation *is*. Second, the mixture is measurable: receiver categories should covary with the action faced—condemnation concentrated at low offers, cooperation at generous sends—which the second-mover representation data can check directly. Third, treatments reach the receiver through the same two levers as the sender (F_c and $\boldsymbol{\kappa}_d$), but nothing forces the two sides to move equally—the asymmetry between first- and second-mover treatment effects that the paper documents.

This closes the model on the forecast-error side. The sender's imagined receiver—the schedule (a, b) or the share s—is retrieved from the *sender's* memory given the sender's description; actual second-mover behavior is retrieved from the *receiver's* memory given a description that includes the action. Nothing aligns the two retrievals, so systematic, context-dependent forecast errors are an equilibrium object, and a treatment that moves the two descriptions differently moves *mean* forecast errors—the paper's treatment-level result. Learning under a stable context is then convergence of the sender's retrieved receiver toward the receiver's retrieved self.

7 Relation to BGLS (2026)

Equation (8) is, structurally, the description-augmented recognition problem of Bordalo et al. (2026) (their eq. 20). The mapping is exact. Our chosen representation $\mathbf{r} = (\boldsymbol{\alpha}, \mathbf{x})$ plays the role of their tuned attention vector α_P; our category prototype $(\mathbf{r}_c, \boldsymbol{\kappa}_c)$ and frequency F_c are their prototype (α_c, κ_c) and recency-weighted frequency, defined exactly as in Section 5: $F_c = \sum_{\tau \in c} \delta^{t-\tau}$ (their footnote 7). Our fidelity term is their description-similarity term, which likewise enters *without* a frequency multiplier: in their reading the present description is prominent as current experience; in the age-zero-trace reading of Section 5 it carries weight $\delta^0 = 1$—the same number for the same reason. Our extreme-value shock is their ϵ_c with precision λ, so retrieval probabilities are their logit (their Proposition 2), and *interference* is the softmax denominator—competition among categories—in both models: fidelity and interference are distinct objects, in their framework as in ours. Two of their results transfer directly: attention is tuned to fit the category, neglecting diagnostic-but-discrepant context (their Proposition 1), which makes categorization sticky in F_c; and a change in a feature's bottom-up salience can re-categorize the problem (their Proposition 7)—their theory of framing is our frame lever. The cue-strength refinement of Section 5 is their contrast weights: they scale similarity feature by feature by contrast σ (description contrast σ_δ, bottom-up; category contrast σ_c, from the variability of past experience) and resolve fidelity versus assimilation exactly—with linear distance, the representation follows the description on a feature if and only if $\sigma_c F_c / \sigma_\delta < 1$ (their eq. 22): the description wins when the category is unfamiliar or the current cue is distinctive.

Two differences are substantive, and they are what the games application adds. First, the attended features. Their features are hedonic and event features of lotteries whose values and probabilities are *known*: attention $\alpha \in [0, 1]$ shades each perceived value relative to its true one, so the bound is a real restriction. Our features are the payoff elements of a strategic problem—own earnings, total surplus, the sharing norm—for which no veridical attention profile exists: behavior identifies only relative weights, so the undistorted case $\rho = \sigma = \mu = 1$ is a statement about ratios (no motive slanted relative to another), and their bound has no analogue here. Second, and more important, beliefs. In their framework perceived probabilities are attention-distortions of an objective $\mathbb{P}$ (their eq. 2). In a one-shot game there is no objective distribution over the counterpart's behavior to distort: the category supplies the belief itself—the imagined receiver (a, b) or s is *prototype content*, the counterpart typical of situations in that category. This is the model's one genuinely new retrieval object, and it is what makes systematic forecast errors an equilibrium possibility. BGLS close by listing strategic behavior as a natural application of problem recognition; the present model is that application.

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A Proofs of Propositions

Throughout, attention weights are strictly positive, $\rho, \sigma, \mu > 0$, and in the ultimatum game $a, b > 0$. Each objective below is strictly concave and differentiable on $[0, 1]$, so the constrained maximizer is unique and equals the unconstrained solution projected onto $[0, 1]$ (the censoring in the propositions); comparative statics are stated at interior optima, where the projection is the identity.

Proof of Proposition 1. With $W(t) = 1$ and $\pi_R = t$, the objective is $u(t) = \rho - \sigma t - \frac{\mu}{2}\left(t - \frac{1}{2}\right)^2$, which is strictly concave ($u'' = -\mu < 0$). The first-order condition $-\sigma - \mu\left(t - \frac{1}{2}\right) = 0$ gives $t^* = \frac{1}{2} - \sigma/\mu$. Since $\sigma/\mu > 0$, $t^* < \frac{1}{2}$, so the upper bound never binds and $t^{DG} = \max\{0, t^*\} < \frac{1}{2}$. The solution is strictly decreasing in σ/μ where interior and constant at zero for $\sigma/\mu \geq \frac{1}{2}$; ρ enters the objective only through the additive constant $\rho W(t) = \rho$, so it leaves the maximizer unchanged. ☐

Proof of Proposition 2. The objective $v(t) = (a + bt)(\rho - \sigma t) - \frac{\mu}{2}\left(t - \frac{1}{2}\right)^2$ has $v'(t) = b\rho - a\sigma - 2b\sigma t - \mu\left(t - \frac{1}{2}\right)$ and $v''(t) = -(2b\sigma + \mu) < 0$: v is strictly concave. The first-order condition gives $t\,(2b\sigma + \mu) = b\rho - a\sigma + \frac{\mu}{2}$, and subtracting $\frac{1}{2}$ from the resulting t yields (4). The censored solution lies in $(0, 1)$ if and only if $\left|b\rho - (a + b)\,\sigma\right| < b\sigma + \mu/2$, half the curvature $2b\sigma + \mu$.

(i) If $t^{DG} = 0$, interiority gives $t^{UG} > 0 = t^{DG}$. If $t^{DG} = \frac{1}{2} - \sigma/\mu > 0$, putting the difference over the common denominator $\mu\,(2b\sigma + \mu)$ gives

$$t^{UG} - t^{DG} \;=\; \frac{b\rho - (a + b)\,\sigma}{2b\sigma + \mu} + \frac{\sigma}{\mu} \;=\; \frac{\mu b\rho + 2b\sigma^2 + \mu\sigma\,(1 - a - b)}{\mu\,(2b\sigma + \mu)},$$

and $a + b \leq 1$ makes every term in the numerator non-negative, with $\mu b\rho > 0$; hence $t^{UG} > t^{DG}$.

(ii) Differentiating (4),

$$\frac{\partial t^{UG}}{\partial \rho} = \frac{b}{2b\sigma + \mu} > 0, \qquad \frac{\partial t^{UG}}{\partial \sigma} = -\frac{(a + b)\,\mu + 2b^2\rho}{(2b\sigma + \mu)^2} < 0, \qquad \frac{\partial t^{UG}}{\partial \mu} = -\frac{t^{UG} - \frac{1}{2}}{2b\sigma + \mu},$$

where the middle expression follows from the quotient rule and is negative unconditionally and the last expression is positive when $t^{UG} < \frac{1}{2}$ and negative when $t^{UG} > \frac{1}{2}$: the norm weight pulls the offer toward the equal split.

(iii) Only the numerator of the second term in (4) depends on a, so $\partial t^{UG}/\partial a = -\sigma/(2b\sigma + \mu) < 0$. The sensitivity $\sigma/(2b\sigma + \mu)$ has derivative $\mu/(2b\sigma + \mu)^2 > 0$ in σ and $-\sigma/(2b\sigma + \mu)^2 < 0$ in μ. ☐

Proof of Proposition 3. The objective $w(t) = \rho\,(1 + rt) - \sigma(1 + r)(1 - s)\,t - \frac{\mu}{2}\,(t - t^e(s))^2$ has $w'(t) = \rho r - \sigma(1 + r)(1 - s) - \mu\,(t - t^e(s))$ and $w''(t) = -\mu < 0$, so the first-order condition gives (6). The censored solution lies in $(0, 1)$ if and only if $-\mu\, t^e(s) < \rho r - \sigma(1 + r)(1 - s) < \mu\big(1 - t^e(s)\big)$. At an interior optimum,

$$\frac{\partial t^{TG}}{\partial \rho} = \frac{r}{\mu} > 0, \qquad \frac{\partial t^{TG}}{\partial s} = \frac{\mathrm{d}t^e}{\mathrm{d}s} + \frac{\sigma(1 + r)}{\mu} > 0, \qquad \frac{\partial t^{TG}}{\partial \sigma} = -\frac{(1 + r)(1 - s)}{\mu} < 0,$$

where $\mathrm{d}t^e/\mathrm{d}s = 2(1 + r)\big[1 + (1 + r)(1 - 2s)\big]^{-2} > 0$ by direct differentiation, and $\partial t^{TG}/\partial \mu = -\big[\rho r - \sigma(1 + r)(1 - s)\big]/\mu^2 = -\big(t^{TG} - t^e(s)\big)/\mu$: the norm weight pulls the send toward the equal-payoff send. As $\mu \to 0$ the objective converges to the affine function $\rho + \big[\rho r - \sigma(1 + r)(1 - s)\big]\,t$, whose maximizer is $t = 1$ if the slope is positive and $t = 0$ if it is negative. For a purely material sender, $\rho = \sigma$, the objective reduces to $\sigma\,\pi_S$ with $\pi_S = 1 + t\,\big[s(1 + r) - 1\big]$, so the slope is $\sigma\big[s(1 + r) - 1\big]$: she sends everything if $s(1 + r) > 1$ and nothing if $s(1 + r) < 1$. The anchor's own censoring is immediate: $t^e(s) \geq 1$ if and only if $(1+r)(1-2s) \leq 0$, i.e., $s \geq \frac{1}{2}$. Finally, if $t^{DG}=0$, interiority gives $t^{TG} > 0 = t^{DG}$; if $t^{DG} = \frac{1}{2} - \sigma/\mu$, then

$$t^{TG} - t^{DG} \;=\; t^e(s) - \tfrac{1}{2} + \frac{\rho r - \sigma(1+r)(1-s) + \sigma}{\mu} \;=\; t^e(s) - \tfrac{1}{2} + \frac{r\left[\rho - \sigma(1-s)\right] + \sigma s}{\mu},$$

which is non-negative if and only if $r\big[\rho - \sigma(1-s)\big] + \sigma s \geq \mu\big(\frac{1}{2} - t^e(s)\big)$. Since $t^e(s) \geq \frac{1}{2}$ exactly when $s \geq r/\big(2(1+r)\big)$, the condition $\rho \geq \sigma(1-s)$ suffices there: the numerator is then a sum of two non-negative terms and the right side is non-positive. □

Proof of Proposition 4. Write $\beta \equiv 1 + (1+r)(1-2s_0)$ and $q(x) \equiv 2(1+r)s_1x^2 + \beta x - 1$. Since $q(0) = -1 < 0$ and $q(x) \to \infty$, q has a unique positive root t^e, and the quadratic's slope at that root is positive: $q'(t^e) = 4(1+r)s_1t^e + \beta > 0$. The objective $w(t) = \rho\,(1+rt) - \sigma(1+r)\,\big(1 - s_0 + s_1 t\big)\,t - \frac{\mu}{2}\,(t-t^e)^2$ has $w'(t) = \rho r - \sigma(1+r)(1-s_0) - 2\sigma(1+r)s_1 t - \mu\,(t - t^e)$ and $w''(t) = -\big(\mu + 2\sigma(1+r)s_1\big) < 0$ for every $s_1 \geq 0$: w is strictly concave. The first-order condition gives $t\,\big(\mu + 2\sigma(1+r)s_1\big) = \mu\,t^e + \rho r - \sigma(1+r)(1-s_0)$, which is (7) and reduces to (6) at $s_1 = 0$. Write $D \equiv \mu + 2\sigma(1+r)s_1$ and $N \equiv \mu\,t^e + \rho r - \sigma(1+r)(1-s_0)$, so that $t^{TG} = N/D$; the censored solution lies in $(0,1)$ if and only if $0 < N < D$, i.e., $-\mu\,t^e < \rho r - \sigma(1+r)(1-s_0) < \mu(1-t^e) + 2\sigma(1+r)s_1$. The comparative statics below hold at an interior optimum.

(i) $\partial t^{TG}/\partial\rho = r/D > 0$; $\partial t^{TG}/\partial\sigma = -(1+r)\big[(1-s_0)D + 2s_1N\big]/D^2 < 0$ since $N > 0$; and $\partial t^{TG}/\partial\mu = \big(t^eD - N\big)/D^2 = -\big(t^{TG} - t^e\big)/D$, so the norm weight pulls the send toward the equal-payoff send.

(ii) Implicit differentiation of $q(t^e) = 0$ gives $\partial t^e/\partial s_0 = 2(1+r)\,t^e/q'(t^e) > 0$, so $\partial t^{TG}/\partial s_0 = \big[\mu\,\partial t^e/\partial s_0 + \sigma(1+r)\big]/D > 0$, and $D \geq \mu$ with equality exactly at $s_1 = 0$ gives the damping.

(iii) Implicit differentiation gives $\partial t^e/\partial s_1 = -2(1+r)\,(t^e)^2/q'(t^e) < 0$, so $\partial t^{TG}/\partial s_1 = \big[\mu\,\partial t^e/\partial s_1 \cdot D - 2\sigma(1+r)N\big]/D^2 = \big[\mu\,\partial t^e/\partial s_1 - 2\sigma(1+r)\,t^{TG}\big]/D < 0$: both terms are negative. □

Proof of Proposition 5. Welfare statements follow from the chain rule $\partial V_g/\partial\theta = W_g' \cdot \partial t^g/\partial\theta$, where the pie's sensitivity to the action is $W_{UG}' = p'(t) = b$ (expected pie $p(t)\cdot 1$) and $W_{TG}' = r$, combined with the action derivatives in the proofs of Propositions 2 and 3: for (i), $\partial V_{UG}/\partial\rho = b \cdot b/(2b\sigma + \mu)$ and $\partial V_{TG}/\partial\rho = r \cdot r/\mu$; for (ii), $\partial V_{UG}/\partial\sigma = b \cdot \partial t^{UG}/\partial\sigma < 0$ and $\partial V_{TG}/\partial\sigma = r \cdot \partial t^{TG}/\partial\sigma < 0$; for (iii), $\partial V_g/\partial\mu = W_g' \cdot \partial t^g/\partial\mu$, and both games have $\partial t^g/\partial\mu$ proportional to $-(t^g - t^e)$, with $t^e = \frac{1}{2}$ in the ultimatum game; for (iv), $\partial V_{UG}/\partial a = 1 + b \cdot \partial t^{UG}/\partial a = 1 - b\sigma/(2b\sigma + \mu)$ and $\partial V_{TG}/\partial s = r \cdot \big[\mathrm{d}t^e/\mathrm{d}s + \sigma(1+r)/\mu\big]$. The counterpart's expected payoff: in the ultimatum game $\partial\big[p(t)t\big]/\partial\sigma = \big(bt + p(t)\big)\,\partial t^{UG}/\partial\sigma < 0$; in the trust game $\partial\big[(1-s_a)(1+r)t\big]/\partial\sigma = (1-s_a)(1+r)\,\partial t^{TG}/\partial\sigma < 0$. The sender's own expected payoff: $\partial\big[p(t)(1-t)\big]/\partial\sigma = \big[b(1-t^{UG}) - p(t^{UG})\big]\,\partial t^{UG}/\partial\sigma$, and $b(1-t) - p(t) = b(1-2t) - a$, so with $\partial t^{UG}/\partial\sigma < 0$ the sender's own expected payoff falls precisely when $b(1-2t^{UG}) > a$; a witness: $a = 1/20$, $b = 3/5$, $\rho = 1/10$, $\sigma = 1$, $\mu = 1$ gives the interior offer $t^{UG} = 0.232$, $b(1-2t^{UG}) = 0.32 > 0.05 = a$, and $\partial\big[p(t)(1-t)\big]/\partial\sigma = -0.04 < 0$. □

Proof of Remark 3. For an actual acceptance schedule $\hat{p}$ with $\hat{p}' > 0$, the chain rule gives $\partial\hat{V}_{UG}/\partial\theta = \hat{p}'(t^{UG}) \cdot \partial t^{UG}/\partial\theta$ for $\theta \in \{\rho, \sigma, \mu\}$, the same signs as in (i)–(iii), while $\partial\hat{V}_{UG}/\partial a = \hat{p}'(t^{UG}) \cdot \partial t^{UG}/\partial a = -\hat{p}'\,\sigma/(2b\sigma + \mu) < 0$, reversing (iv); in the trust game $\hat{V}_{TG} = 1 + r\,t^{TG}$ coincides with V_{TG}, so $\partial\hat{V}_{TG}/\partial s = r\big[\mathrm{d}t^e/\mathrm{d}s + \sigma(1+r)/\mu\big] > 0$ is unchanged. □

Proof of Proposition 6. Conditional on trade, $D_{DG} = 1 - 2t^{DG} = 2\sigma/\mu$ and $D_{UG} = 1 - 2t^{UG} = 2\big[(a+b)\sigma - b\rho\big]/(2b\sigma + \mu)$, whence $\partial D_{UG}/\partial\sigma = 2\big[(a+b)\mu + 2b^2\rho\big]/(2b\sigma + \mu)^2 > 0$, $\partial D_{UG}/\partial\rho = -2b/(2b\sigma + \mu) < 0$, and $\partial D_{UG}/\partial\mu = -D_{UG}/(2b\sigma + \mu)$ by direct differentiation. In the trust game $D_{TG} = (1-t) + s_a(1+r)t - (1-s_a)(1+r)t = 1 - t\big[1 + (1+r)(1-2s_a)\big]$, affine in t with slope $-k$,

$k \equiv 1 + (1+r)(1-2s_a)$, positive iff $s_a < (2+r)/\big(2(1+r)\big)$; since $\partial t^{TG}/\partial\mu = -\big(t^{TG} - t^e(s)\big)/\mu$, $\partial D_{TG}/\partial\mu = -\big(D_{TG} - D_{TG}(t^e(s))\big)/\mu$ with $D_{TG}(t^e(s)) = 1 - k\, t^e(s)$; at $s = s_a$ the definition of the anchor gives $k = 1/t^e(s_a)$, so $D_{TG}(t^e(s_a)) = 0$: the gap at the anchor vanishes exactly under correct beliefs. $\square$

Part II. The Disparity-Based Alternative Model (v1)

This part preserves, in full, the model developed before the one in Part I. The moral motive is outcome-based—a quadratic cost of the payoff disparity—and beliefs are point forecasts of the receiver's attention weights, with *projection* (forecasting the other with one's own weights) as a focal case. The motive triad maps onto Part I as $\omega_o \leftrightarrow \sigma$ (own earnings), $\omega_e \leftrightarrow \rho - \sigma$ (efficiency), and $\omega_d \leftrightarrow \mu$ (norm); the core predictions coincide, and the correspondence is recorded in the appendix cut from the paper (preserved, commented, in the paper source).

II.1 Setup

II.1.1 Games and Payoffs

Two players interact once. Player S is the first mover: the dictator in the Dictator game, the proposer in the Ultimatum game, and the investor in the Trust game. Player R is the second mover: passive in the Dictator game, the receiver in the Ultimatum game, and the trustee in the Trust game. The first mover's endowment is normalized to one, so all actions and material payoffs are shares of that endowment.

- In the *Dictator Game*, S sends $s_D \in [0, 1]$. Payoffs are $(y_S, y_R) = (1 - s_D, s_D)$.
- In the *Ultimatum Game*, S offers $o \in [0, 1]$. If R accepts, payoffs are $(y_S, y_R) = (1 - o, o)$. If R rejects, payoffs are $(y_S, y_R) = (0, 0)$.
- In the *Trust Game*, S sends $s \in [0, 1]$, the amount sent is multiplied by $m > 1$, and R returns $r \in [0, ms]$. Payoffs are $(y_S, y_R) = (1 - s + r, ms - r)$.

II.1.2 Preferences

Preferences are defined over material payoff pairs. Let $p \in \{S, R\}$ denote a player and let $-p$ denote the other player. Player p evaluates an outcome (y_p, y_{-p}) according to

$$U^p(y_p, y_{-p}; \omega^p) = \omega_o^p \, y_p + \omega_e^p \, (y_p + y_{-p}) - \frac{\omega_d^p}{2}(y_p - y_{-p})^2. \tag{II.1}$$

The vector

$$\omega^p = (\omega_o^p, \omega_e^p, \omega_d^p)$$

collects the player's attention weights. The first component, ω_o^p, is attention to own earnings; the second, ω_e^p, is attention to total earnings or efficiency; the third, ω_d^p, is attention to payoff disparity or inequality. We make the following assumptions on weights:

$$\omega_o^p > 0, \qquad \omega_e^p \geq 0, \qquad \omega_d^p > 0.$$

The strict own-earnings weight rules out agents who do not value own material payoffs. The strict disparity weight gives curvature in the transfer problems and therefore uniqueness of the main interior solutions.

Utility is linear in own and in total earnings: all curvature comes from attention to disparity. This is a deliberate modeling choice. Interior sharing in the games below reflects the strength of the disparity motive, not diminishing marginal utility of money, and linearity delivers closed forms and one-parameter belief statistics throughout; Appendix II.B records which results survive when the two earnings terms are evaluated through an arbitrary concave utility index. Only the relative size of the three attention weights matters for choice: multiplying all three weights by the same positive constant multiplies utility by that constant and leaves behavior unchanged. Thus one can impose the normalization

$$\omega_o^p + \omega_e^p + \omega_d^p = 1,$$

without changing the model's choice predictions. The formulas below use unnormalized weights. Comparative statics with unnormalized weights vary one weight holding the other two fixed. If instead weights are restricted to sum to one, an increase in one weight must be paired with decreases in other weights, so the relevant derivative combines the effects reported below.

II.1.3 Beliefs

In the Ultimatum and Trust games, the first mover's optimal action depends on the responder's behavior, which in turn depends on the responder's attention weights. The analysis uses a point-belief benchmark: in game $g \in \{U, T\}$, the first mover behaves as if the responder has the attention-weight vector

$$\hat{\omega}^{R,g} = (\hat{\omega}_o^{R,g}, \hat{\omega}_e^{R,g}, \hat{\omega}_d^{R,g})$$

for sure. When no confusion is possible, the game superscript g is suppressed and the point belief is written simply as $\hat{\omega}^R$.

The full-projection benchmark is the special case in which the first mover believes that the responder has the same weights:

$$\hat{\omega}^{R,g} = \omega^S.$$

Partial projection can be represented as $\hat{\omega}^{R,g} = (1-\rho)\bar{\omega}^g + \rho\,\omega^S$ with $\rho \in [0,1]$, where $\bar{\omega}^g$ is a reference or population mean; remarks in the two projection subsections record which results extend to this intermediate case.

Two tie-breaking conventions apply throughout: an indifferent receiver accepts, and a first mover indifferent between several optimal actions selects the largest.[1]

II.2 Dictator Game

The dictator chooses a transfer $s_D \in [0,1]$. Since total payoff is fixed at one, the efficiency term is constant. The dictator's objective after substituting the material payoffs into utility is

$$D(s_D; \omega^S) = \omega_o^S(1 - s_D) + \omega_e^S - \frac{\omega_d^S}{2}(1 - 2s_D)^2, \tag{II.2}$$

and the derivative of this objective with respect to the transfer s_D is

$$\Phi_D(s_D; \omega^S) = -\omega_o^S + 2\omega_d^S(1 - 2s_D). \tag{II.3}$$

Proofs of all results are in Appendix II.A.

Proposition II.1 (Dictator Transfer)**.** *The optimal transfer is unique:*

[1] Both are knife-edge events: they determine which closed-form branch applies exactly at a regime boundary, and no comparative-statics result depends on them.

$$s_D^* = \max\left\{0,\ \frac{1}{2} - \frac{\kappa^S}{4}\right\}, \qquad \kappa^S = \frac{\omega_o^S}{\omega_d^S},$$

where κ^S is the dictator's selfishness ratio*.*

1. *The transfer is interior — in $(0, 1/2)$ — exactly when $\kappa^S < 2$, and zero otherwise; the dictator never gives more than half.*
2. *At an interior transfer,*

$$\frac{\partial s_D^*}{\partial \omega_o^S} < 0, \qquad \frac{\partial s_D^*}{\partial \omega_e^S} = 0, \qquad \frac{\partial s_D^*}{\partial \omega_d^S} > 0,$$

and these signs hold globally as weak inequalities: s_D^* is nonincreasing in ω_o^S, constant in ω_e^S, and nondecreasing in ω_d^S on the whole parameter space, with strictness exactly at an interior transfer.

Intuition. The dictator trades off own earnings against payoff equality. A marginal unit transferred has the constant own-payoff cost ω_o^S and, as long as $s_D < 1/2$, the disparity benefit $2\omega_d^S(1 - 2s_D)$, which falls linearly as the transfer approaches one half. If the cost exceeds the benefit already at $s_D = 0$ ($\kappa^S \geq 2$), the dictator gives nothing; otherwise the unique interior transfer equates the two forces, which is the closed form above. Greater own-earnings attention raises the cost of giving and lowers the transfer; greater disparity attention raises its benefit and raises the transfer. Efficiency attention is irrelevant because the transfer does not change total surplus.

II.3 Ultimatum Game

II.3.1 Ultimatum Game Receiver

If the receiver accepts offer o, the receiver's payoff is o, the proposer's payoff is $1 - o$, and total payoff is one. The receiver accepts iff

$$A(o; \omega^R) = \omega_o^R\, o + \omega_e^R - \frac{\omega_d^R}{2}(1 - 2o)^2 \geq 0. \tag{II.4}$$

Rejection gives utility zero. Equation (II.4) is the receiver's acceptance value: the receiver accepts exactly when this value is weakly positive.

Proposition II.2 (Ultimatum Minimum Acceptable Offer)**.** *On the interval $o \in [0, 1/2]$, acceptance is governed by a unique minimum acceptable offer $\tau^*(\omega^R) \in [0, 1/2]$:*[2]

$$R \text{ accepts } o \quad \Longleftrightarrow \quad o \geq \tau^*(\omega^R).$$

1. *If $\omega_e^R \geq \frac{\omega_d^R}{2}$, then $\tau^*(\omega^R) = 0$: the receiver accepts every offer.*
2. *If $\omega_e^R < \frac{\omega_d^R}{2}$, the minimum acceptable offer is interior, with closed form*

[2] Restricting attention to $o \leq 1/2$ is without loss for equilibrium behavior. A sufficiently disparity-averse receiver may reject offers close to one, so the acceptance set need not extend to the right of $1/2$ in general; but the proposer never makes such offers, because the accepted-offer objective is strictly concave with maximizer at most $1/2$, and the offer $1/2$ is always accepted (see the proof of Proposition II.3).

$$\tau^*(\omega^R) = \frac{\omega_o^R + 2\omega_d^R - \sqrt{D}}{4\omega_d^R} \in (0, 1/2), \qquad D = (\omega_o^R)^2 + 4\omega_o^R\omega_d^R + 8\omega_d^R\omega_e^R,$$

the smaller root of the acceptance value; the expression under the root is always strictly positive.

3. *At an interior minimum acceptable offer,*

$$\frac{\partial\tau^*}{\partial\omega_o^R} < 0, \qquad \frac{\partial\tau^*}{\partial\omega_e^R} < 0, \qquad \frac{\partial\tau^*}{\partial\omega_d^R} > 0.$$

These signs hold globally as weak inequalities: τ^* is nonincreasing in ω_o^R and ω_e^R and nondecreasing in ω_d^R on the whole parameter space, with strictness exactly at an interior minimum acceptable offer.

Intuition. Rejection gives zero utility. Accepting a low offer has two positive components for the receiver: the receiver obtains own earnings and the pair preserves the unit of surplus. It also has one negative component: the accepted allocation is unequal in favor of the proposer. Raising ω_o^R or ω_e^R increases the value of acceptance relative to rejection, so the receiver is willing to accept lower offers. Raising ω_d^R increases the cost of accepting an unequal offer, so the receiver requires a higher minimum acceptable offer. The zero-offer case occurs when the efficiency value of preserving surplus is already large enough to offset the disparity cost of accepting $o = 0$. The acceptance value is a quadratic in the offer, which is why the minimum acceptable offer — its smaller root — has a closed form.

II.3.2 Ultimatum Game Sender

Let the proposer's belief about the receiver's weights be $\hat\omega^{R,U}$. The believed minimum acceptable offer is $\hat\tau = \tau^*(\hat\omega^{R,U})$, with τ^* given by Proposition II.2. The regime is governed by $\Phi_D(\hat\tau; \omega^S)$, the proposer's marginal value of raising the offer just above the believed minimum acceptable offer, which for every parameter configuration factors as

$$\Phi_D\big(\hat\tau; \omega^S\big) = \omega_d^S\left[\Gamma(\hat\tau) - \kappa^S\right], \qquad \Gamma(\tau) = 2(1 - 2\tau), \tag{II.5}$$

where κ^S is the proposer's selfishness ratio from Proposition II.1 and Γ is a strictly decreasing *toughness index* on $[0, 1/2]$. The regime is therefore determined by a single comparison, $\kappa^S \lesseqgtr \Gamma(\hat\tau)$: the proposer's own weights matter only through κ^S (the efficiency weight not at all), and beliefs matter only through $\hat\tau$.

The offer can be of two kinds. When the acceptance constraint is slack, the proposer offers what he would transfer as a dictator: the offer reflects only his own distributional concerns and is locally independent of beliefs about the receiver. We call such an offer *intrinsic*. When the constraint binds, the proposer offers exactly what he believes the receiver requires: the offer reflects only beliefs about the receiver and is locally independent of the proposer's own weights. We call such an offer *strategic*. As the next proposition shows, the regime comparison $\kappa^S \lesseqgtr \Gamma(\hat\tau)$ determines which case obtains.

Proposition II.3 (Ultimatum Offer)**.** *The proposer offers*

$$o^*(\omega^S, \hat\omega^{R,U}) = \max\{s_D^*(\omega^S), \hat\tau\}. \tag{II.6}$$

1. *If $\kappa^S < \Gamma(\hat{\tau})$, the offer is* intrinsic*: $s^*_D(\omega^S) > \hat{\tau}$ and $o^* = s^*_D(\omega^S) > 0$.*
2. *If $\kappa^S > \Gamma(\hat{\tau})$, then $o^* = \hat{\tau}$. When $\hat{\tau} > 0$ the offer is* strategic*; when $\hat{\tau} = 0$ this is the zero-offer corner, with $s^*_D(\omega^S) = \hat{\tau} = o^* = 0$.*
3. *If $\kappa^S = \Gamma(\hat{\tau})$, then $o^* = s^*_D(\omega^S) = \hat{\tau}$. The offer is continuous at this kink, and comparative statics there are one-sided.*
4. *By Proposition II.2, the toughness index at the believed minimum acceptable offer has the closed form*

$$\Gamma(\hat{\tau}) = \min\left\{2,\ \frac{\sqrt{\hat{D}} - \hat{\omega}^R_o}{\hat{\omega}^R_d}\right\}, \qquad \hat{D} = (\hat{\omega}^R_o)^2 + 4\hat{\omega}^R_o\hat{\omega}^R_d + 8\hat{\omega}^R_d\hat{\omega}^R_e,$$

   with the minimum attained by the first argument exactly when $\hat{\omega}^R_e \geq \hat{\omega}^R_d/2$, the case in which the believed receiver is soft ($\hat{\tau} = 0$). The comparison of items 1–3, $\kappa^S \lesseqgtr \Gamma(\hat{\tau})$, then identifies all cases, and the equilibrium offer, directly from the six weights:

   (a) *if $\kappa^S < \Gamma(\hat{\tau})$, the offer is intrinsic:*

   $$o^* = \frac{1}{2} - \frac{\kappa^S}{4} > 0;$$

   (b) *if $\kappa^S > \Gamma(\hat{\tau})$ and $\hat{\omega}^R_e < \hat{\omega}^R_d/2$, the offer is strategic:*

   $$o^* = \hat{\tau} = \frac{\hat{\omega}^R_o + 2\hat{\omega}^R_d - \sqrt{\hat{D}}}{4\hat{\omega}^R_d} \in (0, 1/2);$$

   (c) *if $\kappa^S \geq 2$ and $\hat{\omega}^R_e \geq \hat{\omega}^R_d/2$, the zero-offer corner obtains: $o^* = 0$;*

   (d) *if $\kappa^S = \Gamma(\hat{\tau})$ with $\hat{\tau} > 0$, the offer is at the kink: $o^* = s^*_D = \hat{\tau}$.*

Intuition. Conditional on acceptance, the proposer's problem is the same as the Dictator problem: the proposer chooses how much of a fixed pie to give away. The difference is the acceptance constraint. The offer is therefore the Dictator transfer *left-censored* at the believed minimum acceptable offer. At the believed minimum acceptable offer, the marginal own-payoff cost of raising the offer is ω^S_o and the marginal disparity benefit is $2\omega^S_d(1 - 2\hat{\tau})$; dividing their difference by ω^S_d yields the regime comparison $\kappa^S \lesseqgtr \Gamma(\hat{\tau})$. If the disparity benefit is larger, the proposer would like to offer more than required for acceptance and the constraint is slack; if it is smaller, the constraint binds and the offer moves one-for-one with the believed minimum acceptable offer.

The censoring structure delivers a single global comparative-statics result: each parameter moves exactly one of the two branches, always in a fixed direction, so the offer is globally monotone and the regime matters only for which effects are locally strict.

Proposition II.4 (Ultimatum Offer: Global Comparative Statics)**.** *The offer o^* is continuous in $(\omega^S, \hat{\omega}^{R,U})$ and* globally *monotone in every parameter, with no regime qualification: at every point where the relevant derivative exists,*

$$\frac{\partial o^*}{\partial \omega^S_o} \leq 0, \qquad \frac{\partial o^*}{\partial \omega^S_e} = 0, \qquad \frac{\partial o^*}{\partial \omega^S_d} \geq 0,$$

$$\frac{\partial o^*}{\partial \hat{\omega}^R_o} \leq 0, \qquad \frac{\partial o^*}{\partial \hat{\omega}^R_e} \leq 0, \qquad \frac{\partial o^*}{\partial \hat{\omega}^R_d} \geq 0,$$

and at the remaining kink points the same weak inequalities hold for both one-sided derivatives. Moreover:

1. (Strictness.) *The own-weight inequalities (ω_o^S, ω_d^S) are strict if and only if the offer is intrinsic ($\kappa^S < \Gamma(\hat{\tau})$); the belief inequalities are strict if and only if the offer is strategic ($\kappa^S > \Gamma(\hat{\tau})$ and $\hat{\tau} > 0$).*

2. (Monotone regime switching.) *The regime gap $\Gamma(\hat{\tau}) - \kappa^S$ is strictly decreasing in ω_o^S, constant in ω_e^S, strictly increasing in ω_d^S; and, whenever $\hat{\tau} > 0$, strictly increasing in $\hat{\omega}_o^R$ and $\hat{\omega}_e^R$ and strictly decreasing in $\hat{\omega}_d^R$. Hence along any one-parameter path the offer changes regime at most once, and the full path of the offer is pinned down. For example, as ω_o^S increases the offer falls strictly until it reaches $\hat{\tau}$ and is constant at $\hat{\tau}$ thereafter; as $\hat{\omega}_d^R$ increases the offer is constant at s_D^* until $\hat{\tau}$ reaches s_D^* and rises strictly, one-for-one with $\hat{\tau}$, thereafter.*

Remark II.1 (Reading the result)*.* All comparative statics follow from the censoring structure of (II.6): the offer is globally monotone in each parameter, the only role of the regime is to determine *which* effects are locally strict, and the regime itself moves monotonically (the regime gap $\Gamma(\hat{\tau}) - \kappa^S$ falls with own greed and believed toughness, rises with own disparity attention and believed materialism or efficiency-mindedness); by the factorization (II.5), all of these movements operate through just two scalars, the selfishness ratio κ^S and the believed minimum acceptable offer $\hat{\tau}$. The censoring structure is also directly useful for estimation: predicted offers follow a Tobit-like specification with latent index s_D^* and stochastic censoring point $\hat{\tau}$.

II.3.3 Ultimatum Game Under Full Projection

Under full projection the proposer forecasts the receiver with his own attention weights, $\hat{\omega}^{R,U} = \omega^S$, so the model delivers predictions as a function of a single weight vector while still allowing beliefs about receiver behavior to matter. (The Dictator game involves no beliefs, so projection leaves it unchanged.) The proposer evaluates the receiver's acceptance problem at his own weights: the believed minimum acceptable offer is $\tau^*(\omega^S)$, characterized by Proposition II.2 evaluated at $\omega^R = \omega^S$ — zero when $\omega_e^S \geq \omega_d^S/2$ and interior otherwise, decreasing in ω_o^S and ω_e^S, increasing in ω_d^S. This object describes the proposer's belief about receiver behavior; the actual receiver's minimum acceptable offer is still $\tau^*(\omega^R)$.

Corollary II.1 (Projected Ultimatum Offer)**.** *Under full projection, $o^P(\omega^S) = \max\{s_D^*(\omega^S), \tau^*(\omega^S)\}$, and Proposition II.3 applies with $\hat{\tau} = \tau^*(\omega^S)$. At an interior offer ($o^P > 0$), the sign pattern is* regime-free*:*

$$\frac{\partial o^P}{\partial \omega_o^S} < 0, \qquad \frac{\partial o^P}{\partial \omega_e^S} \leq 0, \qquad \frac{\partial o^P}{\partial \omega_d^S} > 0,$$

with the ω_e^S inequality strict exactly in the strategic regime ($\kappa^S > \Gamma(\tau^*(\omega^S))$ and $\tau^*(\omega^S) > 0$). At the kink between the two regimes both branches move in the same direction for ω_o^S and ω_d^S, so the strict signs hold there as one-sided derivatives.

Remark II.2 (Identification)*.* Under projection, own-weight comparative statics cannot distinguish the intrinsic from the strategic regime through ω_o^S or ω_d^S: a greedier proposer offers less in both regimes (in the strategic regime because he *projects* greed onto the receiver and therefore believes low offers are accepted). The regimes are separated only by the efficiency weight: ω_e^S is irrelevant for an intrinsic proposer but strictly lowers the offer of a strategic one, because a proposer who values efficiency projects tolerance of unequal-but-efficient allocations onto the receiver.

Remark II.3 (Partial projection). Under partial projection, $\hat{\omega}^{R,U} = (1-\rho)\bar{\omega} + \rho\,\omega^S$ with $\rho \in [0,1]$, the analysis is unchanged with $\hat{\tau} = \tau^*\big((1-\rho)\bar{\omega} + \rho\,\omega^S\big)$: the offer is still $\max\{s_D^*, \hat{\tau}\}$ and the regime comparison is still $\kappa^S \lesseqgtr \Gamma(\hat{\tau})$. Each own-weight effect is the free-beliefs preference channel plus ρ times the corresponding belief channel of Proposition II.4, and for every weight the two channels agree in sign: negative for ω_o^S, negative (through beliefs only) for ω_e^S, positive for ω_d^S. The sign pattern of Corollary II.1 therefore holds for every $\rho \in (0,1]$, and $\rho = 0$ reduces to Proposition II.4.

II.4 Trust Game

II.4.1 Trust Game Receiver

Given sender action s, the receiver chooses $r \in [0, ms]$. Total payoff $1 + (m-1)s$ is fixed from the receiver's perspective. Define the receiver-side payoff gap

$$g_R(r,s) = (ms - r) - (1 - s + r) = (m+1)s - 1 - 2r.$$

The receiver's objective function (after the sender has chosen s) is

$$R(r; s, \omega^R) = \omega_o^R(ms - r) + \omega_e^R\big(1 + (m-1)s\big) - \frac{\omega_d^R}{2} g_R(r,s)^2. \tag{II.7}$$

The derivative of this objective with respect to the amount returned, r, is

$$\Phi_R(r, s; \omega^R) = -\omega_o^R + 2\omega_d^R\big((m+1)s - 1 - 2r\big). \tag{II.8}$$

Proposition II.5 (Trust Return)**.** *For each $s \in [0,1]$, the receiver's return is unique:*

$$r^*(s; \omega^R) = \max\Big\{0,\ \frac{(m+1)s - 1}{2} - k\Big\}, \qquad k = \frac{\omega_o^R}{4\omega_d^R},$$

where k is the receiver's keep premium*: whenever the return is positive, the receiver keeps a final-payoff advantage of exactly $2k$ ($y_R - y_S = 2k$).*

1. (Cutoff and slope.) *Returns are zero at or below the cutoff $\bar{s}_R(\omega^R) = \frac{1+2k}{m+1}$, which is interior exactly when $k < m/2$, and positive above it; positive returns are interior (the corner $r = ms$ never binds) and rise with the send at rate $r_s^* = \frac{m+1}{2} > 1$.*

2. (Comparative statics.) *At an interior return,*

$$\frac{\partial r^*}{\partial \omega_o^R} < 0, \qquad \frac{\partial r^*}{\partial \omega_e^R} = 0, \qquad \frac{\partial r^*}{\partial \omega_d^R} > 0.$$

These signs hold globally as weak inequalities, with strictness exactly at an interior return; the cutoff is increasing in ω_o^R, independent of ω_e^R, and decreasing in ω_d^R.

Intuition. The motive to return comes from disparity attention. Returning lowers the receiver's own material payoff and leaves total surplus fixed, so own-earnings attention pushes toward zero returns and efficiency attention is irrelevant for the return choice. Positive returns begin only when, absent a return, the receiver would be sufficiently ahead that the marginal reduction in the disparity penalty outweighs the marginal own-payoff cost. The cutoff therefore rises with receiver own-earnings attention and falls with receiver disparity attention. In the positive-return region, the marginal return exceeds one because an additional unit sent both increases the receiver's pre-return resources by m and lowers the sender's pre-return resources by one. Since $m > 1$, the receiver must raise the return by more than one unit to restore his keep premium. The marginal return r_s^* is reported because it enters the sender's marginal payoff below: the sender's own-payoff term has marginal effect proportional to $-1 + \hat{r}_s$.

II.4.2 Trust Game Sender

Let the sender's point belief about the receiver's weights in the Trust game be $\hat{\omega}^{R,T}$. By Proposition II.5, the believed receiver is summarized by a single number, the *believed keep premium*, and the believed return strategy is

$$\hat{r}(s) = \max\left\{0,\ \frac{(m+1)s - 1}{2} - \hat{k}\right\}, \qquad \hat{k} = \frac{\hat{\omega}_o^{R,T}}{4\hat{\omega}_d^{R,T}}.$$

Either $\hat{k} \geq m/2$ — the believed receiver never returns, and the sender solves the strictly concave "cautious" problem analyzed below on all of $[0, 1]$ — or the believed cutoff $\hat{s}_R = \frac{1+2\hat{k}}{m+1}$ is interior, with $\hat{r} = 0$ at or below the cutoff and $\hat{r} > 0$ above it. Call $[0, \hat{s}_R]$ the *cautious region* and $(\hat{s}_R, 1]$ the *trusting region*. The names refer to the sender's expectations, not to the size of the send. A cautious sender sends weakly below the cutoff and expects nothing back: any positive amount he sends is motivated intrinsically, by efficiency and equality, exactly as in a Dictator game whose stakes are scaled by the multiplier. A trusting sender deliberately crosses the cutoff and relies on the believed return, and his optimal send is then *full sending*, $s = 1$ (Lemma II.1).

The analysis has three steps. Lemma II.1 establishes the shape of the problem: the objective is strictly concave below the cutoff, its slope jumps *upward* at the cutoff where returns are believed to begin, and it is strictly increasing above the cutoff, so the cutoff itself is never chosen and the sender's choice reduces to a comparison between two plans — the best *cautious* send and *full sending.* Proposition II.6 shows that beliefs about the receiver move only the value of the trusting plan, and only in one direction, so the decision to trust is monotone in beliefs. Proposition II.7 shows how the sender's own weights enter the comparison, through simple payoff differences between the two plans. The trust decision then collapses to a single threshold in the belief statistic $\hat{k}$ (Proposition II.8).

After substituting the believed return strategy, the sender predicts final payoffs

$$\hat{y}_S(s) = 1 - s + \hat{r}(s), \qquad \hat{y}_R(s) = ms - \hat{r}(s),$$

total payoff $q(s) = 1 + (m-1)s$, and payoff gap

$$\hat{d}(s) = \hat{y}_S(s) - \hat{y}_R(s) = 1 - (m+1)s + 2\hat{r}(s).$$

On the trusting region the believed gap is constant, $\hat{d}(s) \equiv -2\hat{k}$: the receiver is believed to defend his keep premium one for one. The sender's objective is

$$T(s; \omega^S, \hat{\omega}^{R,T}) = \omega_o^S\, \hat{y}_S(s) + \omega_e^S\, q(s) - \frac{\omega_d^S}{2} \hat{d}(s)^2, \tag{II.9}$$

and, where $\hat{r}$ is differentiable, its derivative with respect to s is

$$\Phi_T(s; \omega^S, \hat{\omega}^{R,T}) = \omega_o^S(-1 + \hat{r}_s) + \omega_e^S(m-1) + \omega_d^S \hat{d}\big((m+1) - 2\hat{r}_s\big). \tag{II.10}$$

The three terms of (II.10) are the marginal own-payoff effect of sending — a unit sent costs one and is believed to bring back $\hat{r}_s$ — the efficiency gain from the multiplier, and the disparity effect. Lemma II.1 makes the two-plan structure precise.

Lemma II.1 (Two-Branch Structure)**.** *Suppose $\hat{s}_R \in (0,1)$, i.e. $\hat{k} < m/2$. Then:*

1. *On the cautious region, T is a strictly concave quadratic with unconstrained maximizer*

$$\tilde{s}_C = \frac{1}{m+1} + \frac{\omega_e^S(m-1) - \omega_o^S}{\omega_d^S(m+1)^2};$$

the cautious candidate *is its clipped version $s_C = \min\{\max\{\tilde{s}_C, 0\}, \hat{s}_R\}$, and it is unique.*

2. *At $\hat{s}_R$, T has a strictly* convex *kink: the sender's marginal payoff jumps up by $\frac{m+1}{2}\big[\omega_o^S + 4\omega_d^S\hat{k}\big] > 0$.*

3. *On the trusting region, T is affine and strictly increasing, with slope $(m-1)\big(\frac{\omega_o^S}{2} + \omega_e^S\big) > 0$. Hence the unique trusting candidate is full sending, with believed return $\hat{r}(1) = \frac{m}{2} - \hat{k}$ and believed payoffs $\hat{y}_S(1) = \frac{m}{2} - \hat{k}$, $\hat{y}_R(1) = \frac{m}{2} + \hat{k}$.*

4. *Consequently T is never concave on $[0,1]$, the cutoff is never optimal ($s_T^* \neq \hat{s}_R$), and*

$$s_T^* \in \{s_C,\, 1\}:$$

the sender's problem is a comparison of two candidates, the best cautious send and full sending.

Remark II.4 (Trusting is all or nothing). The economics of Lemma II.1 is the following. At the believed cutoff the receiver is, by his own optimality condition, still ahead of the sender. The instant sends cross the cutoff, each extra unit sent is believed to come back at rate $\frac{m+1}{2} > 1$, raising the sender's own payoff and expanding the pie while leaving the believed gap constant at $-2\hat{k}$: all three motives in the sender's marginal payoff point (weakly) toward sending more, so once trusting pays at the margin, it pays all the way to the corner. Trusting is therefore an all-or-nothing proposition: the sender either stays strictly below the believed cutoff or sends everything, the problem is intrinsically bimodal, and no concavity assumption on $[0,1]$ is consistent with the model when the believed cutoff is interior. Appendix II.B shows this is not an artifact of linearity: the all-or-nothing structure holds for any concave earnings utility. All results below work directly with the two-candidate structure of Lemma II.1. Figure II.1 illustrates the two plans and the belief threshold.

Beliefs enter the sender's problem only through the believed return function. Say that a belief $\hat{\omega}'$ is *more pessimistic* than $\hat{\omega}$ if $\hat{\omega}_o' \geq \hat{\omega}_o$ and $\hat{\omega}_d' \leq \hat{\omega}_d$: the believed receiver attends more to own earnings and less to the payoff gap, and by Proposition II.5 returns weakly less at every send. Lower believed returns can only hurt the sender: whenever the believed receiver returns anything he is, at his own optimum, still ahead of the sender, so an extra unit returned raises the sender's own payoff and reduces inequality at the same time. Pessimism therefore lowers the value of every trusting plan, leaves every cautious plan untouched, and can only switch the sender from trusting to caution — never the reverse. The believed efficiency weight is irrelevant throughout, because the return choice reallocates a fixed total. The next proposition records these claims; no assumptions beyond the setup are needed. (And because the trusting send is pinned at $s = 1$ by Lemma II.1, beliefs move only the *decision* to trust, never the size of a trusting send.)

Proposition II.6 (Trust Sender: Monotone Belief Effects)**.** *Fix ω^S and let $\hat{\omega}'$ be more pessimistic than $\hat{\omega}$ (that is, $\hat{\omega}_o' \geq \hat{\omega}_o$ and $\hat{\omega}_d' \leq \hat{\omega}_d$, with $\hat{\omega}_e'$ arbitrary). Then:*

1. (Values.) *$\hat{r}(s;\hat{\omega}') \leq \hat{r}(s;\hat{\omega})$ and $T(s;\hat{\omega}') \leq T(s;\hat{\omega})$ for every s, with equality at every s at which $\hat{r}(s;\hat{\omega}) = 0$. In particular the cautious candidate and its value are unchanged.*

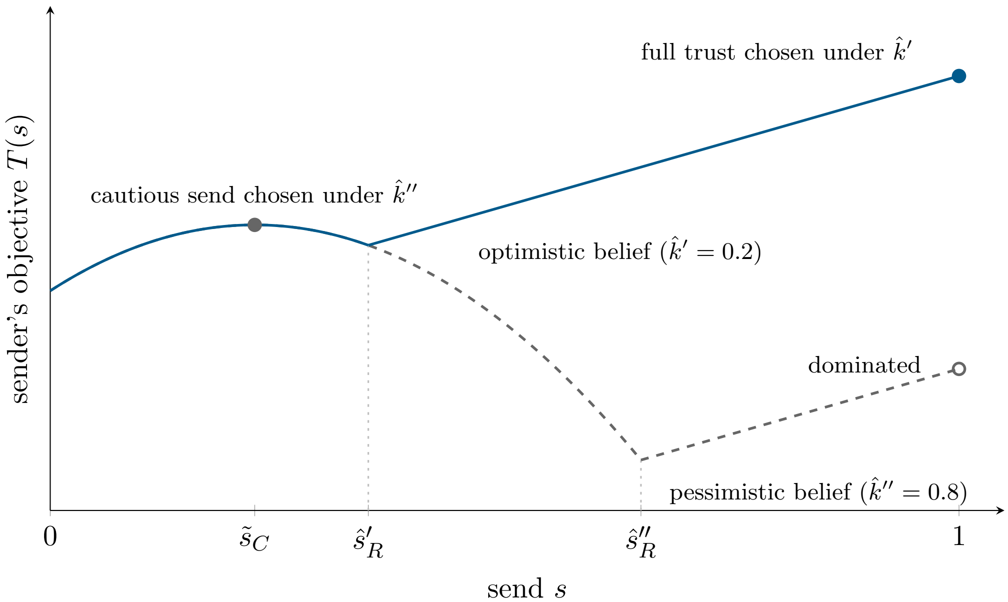

Figure II.1: The sender's two plans. The objective $T(s)$ at $m = 3$, $\omega^S = (1,\, 0.3,\, 1)$, under an optimistic belief ($\hat{k}' = 0.2$, solid) and a pessimistic belief ($\hat{k}'' = 0.8$, dashed). The cautious branch is belief-independent, so the two curves coincide below the lower cutoff $\hat{s}'_R$. Each curve is strictly concave on its cautious region, has a convex kink at its believed cutoff, and rises linearly on its trusting region. Under the pessimistic belief the global optimum is the cautious send $\tilde{s}_C$ (filled circle; the corner $s = 1$ is a dominated local optimum, open circle); under the optimistic belief it is full sending (filled circle). For these preferences the belief threshold is $\bar{k} \approx 0.56$. No send at or near the believed cutoff, or strictly between the cutoff and one, is ever chosen.

2. (Decision.) *If under $\hat{\omega}$ every optimal send is cautious, then under $\hat{\omega}'$ every optimal send is cautious. Equivalently, the willingness to trust is nonincreasing in $\hat{\omega}^R_o$, nondecreasing in $\hat{\omega}^R_d$, and independent of $\hat{\omega}^R_e$.*
3. (Welfare.) *The sender's maximal expected utility is nonincreasing in $\hat{\omega}^R_o$, nondecreasing in $\hat{\omega}^R_d$, and constant in $\hat{\omega}^R_e$.*

The sender's own weights affect both margins. Within the cautious region the problem is a Dictator-like transfer problem scaled by the multiplier, and the cautious send responds to the weights accordingly. The decision between the two plans is governed by the *value of trusting*,

$$G \;=\; \max_{[\hat{s}_R,1]} T \;-\; \max_{[0,\hat{s}_R]} T,$$

the gain from the best trusting plan over the best cautious plan: the sender trusts exactly when $G \geq 0$. Because utility is linear in the three attention weights, G responds to each weight through the difference, between the two plans, of the payoff statistic that the weight multiplies: total payoff for ω^S_e, own payoff for ω^S_o, and the squared payoff gap for ω^S_d.

Proposition II.7 (Trust Sender: Own-Weight Effects)**.**

1. (Cautious send.) *At an interior cautious optimum $s_C \in (0, \hat{s}_R)$,*

$$\boxed{\frac{\partial s_C}{\partial \omega^S_o} < 0, \qquad \frac{\partial s_C}{\partial \omega^S_e} > 0, \qquad \operatorname{sign} \frac{\partial s_C}{\partial \omega^S_d} = \operatorname{sign}\big[1 - (m+1)s_C\big],}$$

and s_C is locally independent of beliefs.

2. (Trust decision.) *G is continuous in ω^S and, since the trusting candidate is full sending (Lemma II.1),*

$$\frac{\partial G}{\partial \omega_e^S} = (m-1)(1-s_C) > 0, \qquad \frac{\partial G}{\partial \omega_o^S} = \Big(\frac{m}{2} - \hat{k}\Big) - (1-s_C),$$
$$\frac{\partial G}{\partial \omega_d^S} = \frac{\big(1-(m+1)s_C\big)^2 - 4\hat{k}^2}{2}.$$

Hence: (i) efficiency attention always *favors trusting — given (ω_o^S, ω_d^S) and beliefs, there is a threshold $\bar{\omega}_e$ such that the sender trusts if and only if $\omega_e^S \geq \bar{\omega}_e$ (with the convention $\bar{\omega}_e = 0$ or $+\infty$ when the decision does not switch); (ii) own-earnings attention favors trusting if and only if full sending delivers a higher believed own payoff than the best cautious send, $\frac{m}{2} - \hat{k} > 1 - s_C$; (iii) disparity attention favors trusting if and only if full sending delivers a more equal believed allocation than the best cautious send, $2\hat{k} < \big|1-(m+1)s_C\big|$.*

Intuition. In the cautious region, sending has a direct own-payoff cost of one unit per unit sent, an efficiency benefit because the amount sent is multiplied by $m > 1$, and a disparity effect whose sign depends on which player is ahead: if the sender is ahead, sending reduces the payoff gap and disparity attention raises sending; if the receiver is already ahead, additional sending widens the gap and disparity attention lowers sending. On the extensive margin, sending more always expands the believed pie; whether it also serves own earnings or equality depends on how much of the expanded pie the receiver is believed to hand back. Efficiency attention is therefore the one motive that favors trusting unconditionally — trusting is, in this model, fundamentally an efficiency phenomenon, disciplined by beliefs about the receiver. Own-earnings and disparity attention favor trusting only conditionally, and the conditions are simple payoff comparisons between the two candidate plans.

II.4.3 The Trust Decision: A Threshold in Believed Selfishness

The two-candidate structure reduces the sender's problem to the comparison of two plans, and the believed receiver enters that comparison through the single scalar $\hat{k}$. The decision to trust is therefore a threshold rule in believed selfishness.

Proposition II.8 (Trust Sender: Threshold Characterization)**.**

1. *If $\hat{k} \geq m/2$, the believed receiver never returns, the sender's objective is strictly concave on $[0,1]$, and $s_T^* = \min\{\max\{\tilde{s}_C, 0\}, 1\}$.*

2. *If $\hat{k} < m/2$, the value gap*

$$G(\hat{k}) = T(1) - T(s_C), \qquad T(1) = \omega_o^S\Big(\frac{m}{2} - \hat{k}\Big) + \omega_e^S m - 2\omega_d^S \hat{k}^2,$$

is continuous and strictly decreasing in $\hat{k}$, so there is a unique threshold $\bar{k} = \bar{k}(\omega^S, m) \leq m/2$, possibly $-\infty$, such that

$$s_T^* = 1 \text{ with positive believed return} \quad \Longleftrightarrow \quad \hat{k} \leq \bar{k}.$$

The threshold moves with the sender's weights according to $\operatorname{sign}(\partial\bar{k}/\partial\theta) = \operatorname{sign}(\partial G/\partial\theta)$, *with the derivatives of Proposition II.7(2); and $\partial G/\partial m > 0$ throughout the no-trust region, so the trust region also expands with the multiplier.*

3. *If $\hat{k} > \bar{k}$, the sender is cautious and, necessarily, unconstrained by the cutoff: $s_T^* = \max\{0, \tilde{s}_C\} < \hat{s}_R$, with believed return zero. In particular no sender ever chooses a send at, or just below, the believed cutoff: predicted sends are bimodal — an interior cautious amount or everything.*

Remark II.5 (Where trust comes from). Full trust hands the sender the believed share $\frac{m}{2} - \hat{k}$ of the multiplied pie, and the sender compares it with the cautious payoff. In the purely selfish limit ($\omega_e^S \to 0$, $\omega_d^S \to 0$) the comparison collapses to own payoffs: $s_C = 0$ and $G \geq 0$ iff $\frac{m}{2} - \hat{k} \geq 1$, i.e. $\bar{k} = \frac{m}{2} - 1$. A purely selfish sender fully trusts whenever the believed final-payoff gap conceded to the receiver, $2\hat{k}$, does not exceed $m - 2$: for $m > 2$ even a selfish sender trusts a believed-fair receiver, while for $m < 2$ he never does.

Intuition. Once returns are believed positive, the receiver raises the return at rate $(m+1)/2$, which keeps the believed payoff gap constant at $2\hat{k}$ as s changes. Disparity therefore no longer disciplines additional sending in the trusting region, while both own payoff and total surplus increase with the send: the best trusting plan is full sending. The trust decision then reduces to a single comparison governed by one belief scalar, $\hat{k}$: optimism (low $\hat{k}$) means the sender expects nearly half of the multiplied pie back, and the threshold $\bar{k}$ rises with the motives that value the trusting outcome — efficiency always, own earnings and equality when the believed receiver is fair enough — and with the size of the pie itself.

Propositions II.6 and II.8 sign the *decision* to trust; the analogue of Proposition II.4 would sign the chosen *send* itself, globally. Because the send jumps at the regime switch, global monotonicity requires the within-regime effect and the jump to point the same way. This alignment holds for exactly two of the model's parameters: beliefs and the efficiency weight.

Proposition II.9 (Global Comparative Statics of the Trust Send)**.** *The optimal send is globally monotone in beliefs and in the efficiency weight:*

$$s_T^* \textit{ is nonincreasing under belief pessimism and nondecreasing in } \omega_e^S$$
$$\big(\textit{nonincreasing in } \hat{\omega}_o^R,\ \textit{nondecreasing in } \hat{\omega}_d^R,\ \textit{constant in } \hat{\omega}_e^R\big).$$

In beliefs, the send is a step function: $s_T^* = 1$ for $\hat{k} \leq \bar{k}$, and the belief-independent cautious send $\max\{0, \tilde{s}_C\}$ *above the threshold. No such global statement holds for the remaining parameters: for ω_d^S and m even the within-regime effect on the cautious send is not uniformly signed, and for ω_o^S the within-regime effects are uniformly nonpositive yet the extensive margin can flip the sign.*

The failure in ω_o^S has, however, an exact structure. Because the trusting candidate is pinned at full sending, the value of the trusting plan is *linear* in the sender's weights, while the value of the cautious plan — a maximum of functions linear in the weights — is convex; the value of trusting G is therefore concave in ω_o^S, and concavity disciplines how greed can move the sender across regimes.

Corollary II.2 (Trust in Greed: An Interval)**.** *Suppose $\hat{s}_R \in (0, 1)$, and fix ω_e^S, ω_d^S, and beliefs.*

1. *G is concave in ω_o^S, so the set of own-earnings weights at which the sender trusts is an interval (possibly empty): as greed rises, the sender switches regime at most twice — into full trust, then back out.*
2. *As $\omega_o^S \to \infty$ the cautious send tends to zero and $\partial G/\partial\omega_o^S \to \left(\frac{m}{2} - \hat{k}\right) - 1$: if $\hat{k} < \frac{m}{2} - 1$, an arbitrarily greedy sender fully trusts; if $\hat{k} > \frac{m}{2} - 1$, he sends nothing.*

Remark II.6 (Greed can increase sending). The failure of global monotonicity in ω_o^S is a testable signature of the two-plan structure rather than a nuisance. Within the cautious regime, own-earnings attention lowers the send; on the extensive margin, it favors trusting whenever full sending yields a higher believed own payoff than the best cautious send, $\frac{m}{2} - \hat{k} > 1 - s_C$ (Proposition II.7). This condition holds under optimistic beliefs, and intrinsic motives relax it further by raising the cautious send that trusting sacrifices: the phenomenon is easiest at large multipliers but is *not* confined to $m > 2$ — at $m = 2$, $\omega_e^S = 0$, $\omega_d^S = 1$, $\hat{k} = 0.15$, the send falls from $\frac{1}{3}$ within caution, jumps to 1 near $\omega_o^S \approx 0.27$, and drops back to caution near $\omega_o^S \approx 3.0$. By Corollary II.2 the pattern is nevertheless tightly disciplined: greed pushes the sender into full trust at most once, pushes him back out unless $\hat{k} \leq \frac{m}{2} - 1$, and the trust region in ω_o^S is always a single interval. The model thus predicts that greed reduces sends among non-trusting senders yet can push senders across the trust threshold — a pattern a monotone model cannot generate.

Remark II.7 (Corner, step, and jump). Two features of these results deserve emphasis. First, within the trusting regime the send is pinned at the corner $s = 1$, so every local comparative static of the sender's action is zero there: all sender-side comparative statics in that regime are *threshold* statics — how close the sender is to the regime boundary — while the equilibrium outcome varies only through the receiver's response $r^*(1; \omega^R)$, which is interior and moves smoothly with the receiver's weights (Proposition II.5). Second, the sender's action is discontinuous in his weights and beliefs — smooth within the cautious regime, constant within the trusting regime, and jumping across the dominated middle region at the switch — yet his value is continuous throughout: each branch value is a maximum of functions continuous in the parameters, and at the switch $G = 0$ the sender is exactly indifferent between the two plans, so the jump discards no utility. This is the generic signature of a bimodal objective. Every other behavioral object in the paper solves a strictly concave problem and therefore varies continuously (the transfer, the minimum acceptable offer, the offer, the return); the convex kink of Lemma II.1 is what makes the trust send the exception.

II.4.4 Trust Game Under Full Projection

Under full projection in the Trust game the sender forecasts the trustee with his own weights, $\hat{\omega}^{R,T} = \omega^S$: the believed keep premium becomes the sender's own selfishness ratio over four, $\hat{k} = \kappa^S/4$, and the Trust prediction depends only on (ω^S, m). Each own weight then operates through two channels: a preference channel (the weight enters the sender's objective directly) and a belief channel (the weight enters the projected receiver's return). For the efficiency weight the belief channel is null, because the return choice does not depend on the receiver's efficiency weight (Proposition II.5); for the own-earnings and disparity weights the two channels resolve unambiguously. Write $e^S = \omega_e^S / \omega_d^S$ for the efficiency ratio, alongside κ^S.

Proposition II.10 (Projected Trust Sender)**.** *Under full projection:*

1. (Threshold.) *There is a threshold $\bar{\kappa}(e^S, m) \in (0, 2m]$, nondecreasing in e^S and in m, such that the sender fully trusts ($s_T^* = 1$ with positive believed return) if and only if $\kappa^S \leq \bar{\kappa}(e^S, m)$; otherwise $s_T^* = \max\{0, \min\{\tilde{s}_C, 1\}\}$ with believed return zero.*

2. (Comparative statics: the decision to trust.) *The full-trust indicator is globally monotone in every primitive:*

$$\boxed{\textit{full trust is} \quad \textit{nonincreasing in } \omega_o^S, \qquad \textit{nondecreasing in } \omega_e^S,\ \omega_d^S,\ \textit{and } m.}$$

In particular the ambiguity of the own-earnings effect in Proposition II.9 is resolved: under projection, greed both lowers the appetite for the trusting gamble and poisons the belief about *the receiver, and the net effect is unambiguously against trusting.*

3. (Comparative statics: the cautious send.) *In the cautious regime ($\kappa^S > \bar\kappa$) with an interior send $s^*_T = \tilde s_C \in (0, \hat s_R)$, the belief channel is locally null and*

$$\frac{\partial s^*_T}{\partial \omega^S_o} < 0, \qquad \frac{\partial s^*_T}{\partial \omega^S_e} > 0, \qquad \operatorname{sign} \frac{\partial s^*_T}{\partial \omega^S_d} = \operatorname{sign}\bigl[1 - (m+1)s^*_T\bigr].$$

4. (Closed form.) *In the region where the cautious candidate is interior and unconstrained, the trust condition $\kappa^S \le \bar\kappa(e^S, m)$ is the positivity region of the explicit concave quadratic*

$$\tilde G(\kappa^S, e^S) = \frac{m(m-1)}{2(m+1)}\,\kappa^S + \frac{m(m-1)}{m+1}\,e^S - \frac{3}{8}\,(\kappa^S)^2 - \frac{\bigl(e^S(m-1) - \kappa^S\bigr)^2}{2(m+1)^2},$$

and at $e^S = 0$ the threshold is in closed form:

$$\bar\kappa(0, m) = \frac{4m(m-1)(m+1)}{3(m+1)^2 + 4}.$$

5. (Global comparative statics: the send.) *Combining the two margins, the optimal send itself is globally monotone in ω^S_o and ω^S_e:*

$$s^*_T \textit{ is nonincreasing in } \omega^S_o \quad \textit{and} \quad \textit{nondecreasing in } \omega^S_e,$$

while for ω^S_d and m the effect on the cautious send is not uniformly signed and no global statement holds.

Intuition. Projection turns the sender's belief into a mirror: a sender who attends to own earnings expects the receiver to do the same and keep a premium $2\hat k = \kappa^S/2$ of the multiplied pie. Greed therefore cuts against trusting twice — it raises the standard the trusting gamble must meet, and it worsens the projected terms of that gamble — and the two channels never conflict. Efficiency attention favors trusting twice over: it values the multiplied pie directly and, because the receiver's return choice does not change total surplus, projecting it onto the receiver costs nothing. Disparity attention makes the sender expect equalizing returns and simultaneously makes the near-equal full-trust allocation attractive. This is why, under projection, the Trust game delivers the cleanest possible sign pattern on the extensive margin: $(-, +, +)$ in the three own weights, $+$ in the multiplier.

Remark II.8 (Partial projection). Partial projection, $\hat\omega^{R,T} = (1-\rho)\bar\omega + \rho\,\omega^S$, scales the belief channel by ρ without changing its sign: greater ω^S_o still makes the projected belief more pessimistic and greater ω^S_d more optimistic, so by Proposition II.6 the belief channel of each own weight retains its full-projection direction for every $\rho > 0$. The efficiency-weight monotonicity holds verbatim for every $\rho \in [0, 1]$, because the return choice never depends on the believed efficiency weight. The unambiguous signs for ω^S_o and ω^S_d in Proposition II.10, by contrast, do rely on full projection: for $\rho < 1$ the preference channel can dominate at parameter values where it opposes the belief channel, so the clean monotonicity of the trust decision in those two weights is a genuine full-projection result.

II.5 The Trust Game with a Repayment Norm

This section presents, self-contained and in full, the Trust game in which fair conduct for the receiver is *repayment*: give the sender back what he invested, and keep the surplus.

Recall the game. The sender is endowed with one unit, the receiver with nothing. The sender chooses how much to send, $s \in [0,1]$; the amount sent is multiplied by $m > 1$; the receiver then chooses how much to return, $r \in [0, ms]$. Material payoffs are

$$y_S = 1 - s + r, \qquad y_R = ms - r,$$

and total payoff $q(s) = 1 + (m-1)s$ is fixed once the send is chosen: the send creates surplus, the return only divides it. Players evaluate outcomes with the attention-weighted utility of Section II.1, attaching weight $\omega_o > 0$ to own earnings, $\omega_e \geq 0$ to total earnings, and $\omega_d > 0$ to a disparity penalty.

What the receiver's disparity penalty targets is the departure from the gap-benchmark model of Section II.4. A receiver who returns less than s leaves the sender below his endowment: he has, in effect, kept part of what he was entrusted with. A receiver who returns exactly s makes the sender whole and keeps the entire multiplication surplus $(m-1)s$ for himself. The *repayment norm* takes this boundary as the standard of fair conduct — debts are repaid, windfalls are kept — and the receiver minds falling short of it:

$$U^{R,T} = \omega_o^R (ms - r) + \omega_e^R \, q(s) - \frac{\omega_d^R}{2} \big(s - r\big)^2. \tag{II.11}$$

Unlike the gap-benchmark receiver, who compares final earnings, this receiver compares his action to a reference action: he feels no obligation to share the multiplication surplus, only to give back the stake. Everything else is as in the gap-benchmark model: the sender's disparity concern targets the final payoff gap, and the sender holds a point belief $\hat{\omega}^{R,T}$ about the receiver's weights.

II.5.1 Receiver

Fix a send $s > 0$. Returning is a pure transfer: each unit returned lowers the receiver's own earnings one for one and leaves total earnings $q(s)$ unchanged, so own-earnings attention pushes toward returning nothing and efficiency attention is silent about the return. The one motive to return is the norm: at $r < s$ the receiver is short of full repayment by $s - r$, and the marginal relief from shrinking that debt, $\omega_d^R(s - r)$, is weighed against the constant marginal own-earnings cost ω_o^R.

Proposition II.11 (Repayment Returns)**.** *For each $s \in [0,1]$ the receiver's return is unique:*

$$r^*(s; \omega^R) = \max\big\{0, \; s - c\big\}, \qquad c = \frac{\omega_o^R}{\omega_d^R},$$

where c is the receiver's norm shortfall*.*

1. (Cutoff and slope.) *Returns are zero at or below the cutoff, which* equals *the shortfall, $\bar{s}_I = c$ (interior exactly when $c < 1$), and positive above it; positive returns are interior — the receiver never fully repays — and the marginal return is exactly one: past the cutoff, each additional unit sent is repaid in full.*

2. (Comparative statics.) *At an interior return,*

$$\frac{\partial r^*}{\partial \omega_o^R} < 0, \qquad \frac{\partial r^*}{\partial \omega_e^R} = 0, \qquad \frac{\partial r^*}{\partial \omega_d^R} > 0.$$

These signs hold globally as weak inequalities, with strictness exactly at an interior return; the cutoff is increasing in ω_o^R, independent of ω_e^R, and decreasing in ω_d^R.

Intuition. The receiver behaves like a debtor with a fixed temptation: he would like to pocket $c = \omega_o^R/\omega_d^R$ of the debt. For small investments ($s \leq c$) the temptation swallows the whole debt and nothing comes back; once s exceeds c he returns $s - c$, holding his shortfall from the norm constant — so every marginal unit invested is repaid one for one. More selfish receivers (higher ω_o^R, lower ω_d^R) have larger shortfalls: they return less at every investment and start returning later, and receivers with $c \geq 1$ never return anything. Contrast the gap-based receiver of Section II.4, who starts returning only once he is strictly ahead of the sender and then returns at marginal rate $\frac{m+1}{2} > 1$: the repayment receiver is pinned to his conduct standard instead, and his returns can begin while the sender is still ahead.

II.5.2 Sender

The sender moves first and behaves as if the receiver's weights are $\hat{\omega}^{R,T}$ for sure. The believed receiver is summarized by one number, the *believed shortfall* $\hat{c} = \hat{\omega}_o^R/\hat{\omega}_d^R$: by Proposition II.11, the sender expects the return $\hat{r}(s) = \max\{0, s - \hat{c}\}$, zero at or below the believed cutoff $\hat{c}$ and repaying each marginal unit in full above it. Call $[0, \hat{c}]$ the *cautious region* and $(\hat{c}, 1]$ the *trusting region*: a cautious sender expects nothing back, a trusting sender deliberately crosses the cutoff and relies on the believed repayment. Assume $\hat{c} < 1$, so that the believed cutoff is interior; otherwise the believed receiver never returns and the sender solves the strictly concave cautious problem below on all of $[0, 1]$.

Substituting the believed return into the sender's utility gives believed payoffs $\hat{y}_S(s) = 1-s+\hat{r}(s)$ and $\hat{y}_R(s) = ms - \hat{r}(s)$, believed gap $\hat{d}(s) = \hat{y}_S(s) - \hat{y}_R(s)$, and the objective

$$T(s) = \omega_o^S\, \hat{y}_S(s) + \omega_e^S\, q(s) - \frac{\omega_d^S}{2}\, \hat{d}(s)^2. \tag{II.12}$$

On the cautious region the sender expects nothing back, so his motives are intrinsic — those of a dictator whose transfer is multiplied: sending costs one per unit, expands the pie by $m - 1$ per unit, and moves the payoff gap $1 - (m + 1)s$. The objective there is the belief-free strictly concave quadratic

$$T_0(s) = \omega_o^S(1 - s) + \omega_e^S q(s) - \frac{\omega_d^S}{2}\big(1 - (m + 1)s\big)^2,$$

with unconstrained maximizer

$$\tilde{s}_C = \frac{1}{m+1} + \frac{\omega_e^S(m-1) - \omega_o^S}{\omega_d^S(m+1)^2},$$

and the *cautious candidate* is the clipped version $s_C = \min\{\max\{\tilde{s}_C, 0\}, \hat{c}\}$.

On the trusting region the believed payoffs are

$$\hat{y}_S(s) = 1 - \hat{c}, \qquad \hat{y}_R(s) = (m-1)s + \hat{c}, \qquad \hat{d}(s) = 1 - 2\hat{c} - (m-1)s.$$

Because the believed receiver repays each marginal unit exactly, the sender's believed own payoff is *flat* in the send: once he has decided to trust, he expects to end with his endowment less the shortfall $\hat{c}$, no matter how much he sends. Trusting is believed to cost a fixed fee $\hat{c}$ — the tribute to the receiver's temptation — and, past that fee, the size of the send is a decision about *other people's money*: each extra unit adds $m-1$ to the pie, all of it in the receiver's pocket. Own-earnings attention is therefore indifferent about the size of a trusting send; efficiency attention pushes it up; and disparity attention pulls it back once the receiver moves ahead, since the believed gap falls at rate $m-1$. The trusting branch is thus strictly concave, and it peaks where the believed gap equals $-\omega_e^S/\omega_d^S$: a sender without efficiency attention targets believed *equality of final payoffs*, and efficiency attention pushes the target into the receiver's favor. This is the central difference from the gap-benchmark model, where the believed gap is constant on the trusting region, the trusting branch is strictly increasing, and the only trusting candidate is full sending.

The sender's problem therefore reduces to comparing two plans: the best cautious send and the best trusting send. Write G for the *value of trusting* — the maximum of T over the trusting region minus its maximum over the cautious region — so that the sender trusts exactly when $G \geq 0$.

Lemma II.2 (Two Concave Branches)**.** *Let $\hat{c} < 1$. Then:*

1. *On the cautious region, $T = T_0$: strictly concave, with cautious candidate s_C.*

2. *On the trusting region, T is a strictly concave quadratic with unconstrained maximizer*

$$\tilde{s}_I = \frac{1 - 2\hat{c} + \omega_e^S/\omega_d^S}{m-1}.$$

3. *At $\hat{c}$ the marginal payoff jumps by $\omega_o^S - 2\omega_d^S\big(1-(m+1)\hat{c}\big)$: writing $\kappa^S = \omega_o^S/\omega_d^S$, the kink is strictly convex if $\kappa^S > 2\big(1-(m+1)\hat{c}\big)$ and strictly concave if the inequality is reversed — in which case T is strictly concave on all of $[0,1]$.*

4. *The cutoff itself is never optimal, and $s_T^* \in \{s_C, \min\{\tilde{s}_I, 1\}\}$ whenever $\tilde{s}_I > \hat{c}$; otherwise every optimal send is cautious.*

Unlike in the gap-benchmark model, the kink need not be convex: a norm-based receiver can be believed to start returning while the sender is still ahead ($1-(m+1)\hat{c} > 0$), where the onset of returns *widens* the believed gap and hurts an inequality-averse sender. When the kink is concave the whole problem is single-peaked and the policy continuous in weights and beliefs; otherwise the two-plan comparison and the jump in the policy survive.

Proposition II.12 (Equilibrium Behavior under the Repayment Norm)**.** *Let $\hat{c} < 1$.*

1. (Equilibrium outcomes.) *The optimal send is either the cautious candidate s_C — met, under correct beliefs, by zero return — or the trusting send $\min\{\tilde{s}_I, 1\}$, met by the positive return $s_T^* - c$. On an open set of primitives the optimum is the* interior *trusting send $s_T^* = \tilde{s}_I \in (\hat{c}, 1)$, so under correct beliefs the model generates equilibrium outcomes in which both players' actions are interior.*[3]

2. (Intensive margin.) *At an interior trusting optimum,*

$$\boxed{\begin{gathered}\frac{\partial s_T^*}{\partial \omega_o^S} = 0, \qquad \frac{\partial s_T^*}{\partial \omega_e^S} = \frac{1}{(m-1)\omega_d^S} > 0, \qquad \frac{\partial s_T^*}{\partial \omega_d^S} = -\frac{\omega_e^S}{(m-1)(\omega_d^S)^2} \leq 0, \\ \frac{\partial s_T^*}{\partial \hat{c}} = -\frac{2}{m-1} < 0, \qquad \frac{\partial s_T^*}{\partial m} = -\frac{\tilde{s}_I}{m-1} < 0.\end{gathered}}$$

[3]For example, at $m = 3$, $\omega^S = (1, 0.5, 1)$, $\hat{c} = 0.3$: $s_T^* = 0.45$ and the return is 0.15.

3. (Decision.) *Wherever* $G \le 0$*, the value of trusting is strictly decreasing in* ω_o^S *and* $\hat{c}$ *and strictly increasing in* ω_e^S*, so the trust decision is characterized by a threshold in each parameter. In particular greed is unambiguously against trusting: full repayment caps the trusting own payoff at* $1-\hat{c}$*, always below the cautious own payoff* $1-s_C$*.*

4. (Global comparative statics.)

$$\boxed{\begin{array}{c} s_T^* \textit{ is globally nonincreasing in } \omega_o^S \textit{ and in } \hat{c}, \textit{ and globally nondecreasing in } \omega_e^S \\ \big(\textit{nonincreasing in } \hat{\omega}_o^R, \textit{ nondecreasing in } \hat{\omega}_d^R, \textit{ constant in } \hat{\omega}_e^R\big). \end{array}}$$

No global statement holds for ω_d^S *or* m*.*

Intuition. Every comparative static flows from the flat believed own payoff. Greed: own-earnings attention values the cautious plan at $1-s_C$ and any trusting plan at $1-\hat{c} < 1-s_C$, so it lowers the cautious send, leaves the trusting send untouched, and votes against trusting — sends fall in ω_o^S everywhere. Efficiency: the trusting pie is larger and $\tilde{s}_I$ rises with ω_e^S, so efficiency attention raises sends everywhere. Pessimism: a higher believed shortfall $\hat{c}$ raises the fee to trusting, shrinks the trusting send, and leaves the cautious plan untouched, so sends fall. Only disparity attention remains ambiguous, for the same reason as in the gap-benchmark model: whether sending narrows or widens the believed gap depends on which side of equality the sender starts from, and this is a fact about the sender's own motive, not about the receiver.

Remark II.9 (Greed monotonicity restored)*.* The gap-benchmark model's signature non-monotonicity — greed can push senders across the trust threshold (Proposition II.9) — disappears under the repayment norm. The two receiver benchmarks are thereby separable in data: the gap benchmark predicts that greed pushes some favorably-believing senders into full trust, while the repayment norm predicts a uniformly negative relation between greed and sends.

II.5.3 The Repayment Norm Under Full Projection

Under full projection the sender forecasts the receiver with his own weights, so the believed shortfall is the sender's own selfishness ratio, $\hat{c} = \kappa^S = \omega_o^S/\omega_d^S$, and the efficiency weight again has no belief channel (Proposition II.11). The trusting maximizer becomes

$$\tilde{s}_I = \frac{1-2\kappa^S + e^S}{m-1},$$

and the one margin that was flat under free beliefs comes alive: greed does not change what a trusting sender expects to keep at the margin — that is still exact repayment — but it poisons his belief about the shortfall he must write off, so the trusting send now strictly falls in ω_o^S.

Proposition II.13 (Projected Repayment Sender)**.** *Under full projection:*

1. (Regimes and threshold.) *If* $\kappa^S \ge 1$*, the projected receiver never returns,* T *is strictly concave on* $[0,1]$*, and* $s_T^* = \min\{\max\{\tilde{s}_C, 0\}, 1\}$*. If* $\kappa^S < 1$*, there is a threshold* $\bar{\kappa}_I(e^S, m)$*, nondecreasing in* e^S*, such that optimal sends are trusting if and only if* $\kappa^S \le \bar{\kappa}_I$*.*

2. (Intensive margin.) *At an interior trusting optimum,*

$$\boxed{\frac{\partial s_T^*}{\partial \omega_o^S} = -\frac{2}{(m-1)\omega_d^S} < 0, \qquad \frac{\partial s_T^*}{\partial \omega_e^S} = \frac{1}{(m-1)\omega_d^S} > 0, \qquad \operatorname{sign} \frac{\partial s_T^*}{\partial \omega_d^S} = \operatorname{sign}\big[2\kappa^S - e^S\big].}$$

The greed effect is pure belief channel; interior trusting optima exist on an open set of primitives for every $m > 1$.[4]

3. (Global comparative statics.)

$$\text{for every } m > 1: \quad s_T^* \text{ is globally nonincreasing in } \omega_o^S \text{ and globally nondecreasing in } \omega_e^S.$$

The free-belief greed monotonicity of Proposition II.12 is reinforced: both channels now work against sending. For ω_d^S and m no global statement holds.

II.6 Summary of Comparative Statics

This section collects the model's comparative statics for the Dictator game, the Ultimatum game, and the two variants of the Trust game: the *gap benchmark* of Section II.4, where the receiver's disparity penalty targets the final payoff gap $y_R - y_S$, and the *repayment norm* of Section II.5, where it targets the shortfall from returning the investment. Table II.1 reports local signs under free beliefs, Table II.2 under full projection, and Table II.3 compares the *global* comparative statics of the trust send across the two variants.

[4]For example, at $m = 3$, $\omega^S = (0.3, 0.5, 1)$: $\kappa^S = 0.3$, $s_T^* = \tilde{s}_I = 0.45$, with believed return 0.15.

Table II.1: Comparative statics under free beliefs

Panel A: Second-mover behavior

Object	ω_o^R	ω_e^R	ω_d^R
Ultimatum minimum acceptable offer τ^*	$-$	$-$	$+$
Trust return r^* (either benchmark)	$-$	0	$+$

Panel B: First-mover behavior

	Own weights			Believed receiver weights		
Object	ω_o^S	ω_e^S	ω_d^S	$\hat{\omega}_o^R$	$\hat{\omega}_e^R$	$\hat{\omega}_d^R$
Dictator transfer s_D^*	$-$	0	$+$	NA	NA	NA
Ultimatum offer o^*	$-^{(a)}$	0	$+^{(a)}$	$-^{(b)}$	$-^{(b)}$	$+^{(b)}$
Trust cautious send s_C (either benchmark)	$-$	$+$	$\text{sign}[1-(m+1)s_C]$	0	0	0
Trusting decision — gap benchmark	$\pm^{(c)}$	$+$	$\pm^{(d)}$	$-$	0	$+$
Trusting decision — repayment norm	$-$	$+$	$\pm$	$-$	0	$+$
Interior trusting send $\tilde{s}_I$ — repayment norm	0	$+$	$-^{(e)}$	$-$	0	$+$

Notes. Signs are local effects at interior solutions; 0 denotes a locally zero effect, NA a parameter that does not enter the problem, and $\pm$ a conditional sign. *Trust labels:* the *cautious send* s_C is the optimal send of a sender who expects no return — the same belief-free problem in both variants (Lemmas II.1 and II.2); the *trusting decision* is whether the optimal send crosses the believed return cutoff; a trusting sender sends everything under the gap benchmark, while under the repayment norm his send is typically the *interior trusting send* $\tilde{s}_I$. The Panel A signs and the Dictator and Ultimatum-offer signs in Panel B also hold globally as weak inequalities (Propositions II.2, II.5, II.11, II.1, and II.4); the two Trust benchmarks share the receiver signs but differ in the marginal return to the send, $\frac{m+1}{2}$ (gap) versus 1 (repayment) in the positive-return region. For the Ultimatum offer, $^{(a)}$ strict iff the offer is intrinsic ($\kappa^S < \Gamma(\hat{\tau})$) and $^{(b)}$ strict iff the offer is strategic ($\kappa^S > \Gamma(\hat{\tau})$ and $\hat{\tau} > 0$). Gap-benchmark decision (Propositions II.6 and II.8): $^{(c)}$ positive iff full trust yields a higher believed own payoff than the cautious send, $\frac{m}{2} - \hat{k} > 1 - s_C$ (the trust region in ω_o^S is an interval, Corollary II.2), and $^{(d)}$ positive iff full trust yields a more equal believed allocation. Repayment rows: Proposition II.12; belief signs operate through the believed shortfall $\hat{c} = \hat{\omega}_o^R/\hat{\omega}_d^R$; $^{(e)}$ strict iff $\omega_e^S > 0$.

Table II.2: Comparative statics under full projection

Object	ω_o^S	ω_e^S	ω_d^S
Dictator transfer s_D^*	$-$	0	$+$
Ultimatum offer o^P	$-$	$-^{(a)}$	$+$
Trust cautious send s_C (either benchmark)	$-$	$+$	$\text{sign}[1-(m+1)s_C]$
Full-trust decision — gap benchmark	$-$	$+$	$+$
Interior trusting send $\tilde{s}_I$ — repayment norm	$-$	$+$	$\pm$

Notes. Projection ties the believed weights to the sender's own, so each sign combines the preference and belief channels. The Dictator game involves no beliefs and is unchanged. Ultimatum signs are for interior offers (Corollary II.1); $^{(a)}$ strict exactly in the strategic regime, zero in the intrinsic regime. Gap-benchmark rows: Proposition II.10; the full-trust decision is also nondecreasing in the multiplier m. Repayment row: Proposition II.13; the ω_d^S sign is $\text{sign}[2\kappa^S - e^S]$. Combining margins, under projection the send is globally nonincreasing in ω_o^S and nondecreasing in ω_e^S in both Trust variants.

Table II.3: Global comparative statics of the Trust send

Parameter	Free beliefs: Gap benchmark	Free beliefs: Repayment norm	Full projection: Gap benchmark	Full projection: Repayment norm
ω_o^S	NM$^{(a)}$	–	–	–
ω_e^S	+	+	+	+
ω_d^S	NM	NM	NM	NM
Belief pessimism	–	–	(beliefs tied to own weights)	
Interior trusting send	never	open region	never	open region

Notes. Entries are global weak signs of the optimal send s_T^*, for general $m > 1$; NM = not globally monotone. Gap benchmark: the receiver's disparity penalty targets the final payoff gap (Section II.4); repayment norm: it targets the shortfall from returning the investment (Section II.5). Belief pessimism: higher $\hat{\omega}_o^R$, lower $\hat{\omega}_d^R$. $^{(a)}$ The failure has an exact structure: the trust region in ω_o^S is an interval, and extreme greed ends in full trust iff $\hat{k} < \frac{m}{2} - 1$ (Corollary II.2); the failure is exhibited for $m \geq 2$. Free-belief columns: Propositions II.9 and II.12; projection columns: Propositions II.10 and II.13. The last row records whether an equilibrium outcome with an interior send and an interior return exists (under correct beliefs in the free-belief columns; under projection, when the projected belief is correct): under the gap benchmark a trusting sender always sends everything, while under the repayment norm interior trusting sends arise on an open set of primitives. Several entries extend to concave earnings utility — the ω_e^S row in every column, the gap free-belief column, and the free-belief interior-trusting-send entries — while the repayment ω_o^S sign is a linear-utility result (Appendix II.B).

II.A Proofs

Proof of Proposition II.1

D is a strictly concave quadratic ($D_{ss} = -4\omega_d^S < 0$) with $\Phi_D(s_D) = -\omega_o^S + 2\omega_d^S(1-2s_D)$ strictly negative for $s_D \geq 1/2$, so no maximizer lies weakly above $1/2$. The unconstrained first-order condition solves to $\frac{1}{2} - \frac{\kappa^S}{4}$, positive exactly when $\Phi_D(0) = 2\omega_d^S - \omega_o^S > 0$, i.e. $\kappa^S < 2$; otherwise the maximum is the corner $s_D = 0$. The local comparative statics follow by differentiating the closed form. The global weak versions follow because Φ_D is pointwise nonincreasing in ω_o^S and nondecreasing in ω_d^S on $[0, 1/2)$, and a pointwise shift of the marginal payoff moves the maximizer of a strictly concave objective weakly in the same direction, including at the corner $s_D^* = 0$; ω_e^S does not enter Φ_D. □

Proof of Proposition II.2

For $o < 1/2$, $A_o(o;\omega^R) = \omega_o^R + 2\omega_d^R(1-2o) > 0$, and $A(1/2;\omega^R) = \omega_o^R/2 + \omega_e^R > 0$: on $[0,1/2]$ the acceptance value is strictly increasing and strictly positive by $o = 1/2$, so it crosses zero at most once. Since $A(0;\omega^R) = \omega_e^R - \omega_d^R/2$, the minimum acceptable offer is zero exactly when $\omega_e^R \geq \omega_d^R/2$. Otherwise $A(\tau;\omega^R) = 0$ is the quadratic $2\omega_d^R\tau^2 - (\omega_o^R + 2\omega_d^R)\tau + (\omega_d^R/2 - \omega_e^R) = 0$, whose discriminant simplifies to

$$(\omega_o^R + 2\omega_d^R)^2 - 8\omega_d^R\left(\frac{\omega_d^R}{2} - \omega_e^R\right) = (\omega_o^R)^2 + 4\omega_o^R\omega_d^R + 8\omega_d^R\omega_e^R = D > 0;$$

the quadratic is positive at $\tau = 0$ and negative at $\tau = 1/2$ (it equals $-\omega_o^R/2 - \omega_e^R$ there), so its smaller root is the unique crossing in $(0, 1/2)$, as displayed.

At an interior minimum acceptable offer, the implicit-function theorem applied to $A(\tau;\omega^R) = 0$ gives

$$\frac{\partial\tau^*}{\partial\omega_o^R} = -\frac{\tau^*}{A_o(\tau^*;\omega^R)} < 0, \qquad \frac{\partial\tau^*}{\partial\omega_e^R} = -\frac{1}{A_o(\tau^*;\omega^R)} < 0, \qquad \frac{\partial\tau^*}{\partial\omega_d^R} = \frac{(1-2\tau^*)^2}{2A_o(\tau^*;\omega^R)} > 0.$$

For the global statement: $A(o;\omega^R)$ is pointwise nondecreasing in ω_o^R and ω_e^R and nonincreasing in ω_d^R; since A is strictly increasing on $[0, 1/2]$ and τ^* is its unique crossing point (or the corner 0), a pointwise upward shift of the acceptance value weakly lowers τ^* and a downward shift weakly raises it. □

Proof of Proposition II.3

Offers below $\hat{\tau}$ are rejected and yield utility zero. Offers at or above $\hat{\tau}$ (up to $1/2$) are accepted and yield

$$W(o;\omega^S) = \omega_o^S(1-o) + \omega_e^S - \frac{\omega_d^S}{2}(1-2o)^2,$$

which is the same objective as (II.2) with o in place of s_D. By Proposition II.1, W is strictly concave and maximized at $s_D^*(\omega^S) \in [0, 1/2]$. Offers above $1/2$ are never optimal: if accepted they yield $W(o) < W(1/2)$ by strict concavity, and if rejected they yield $0 < W(1/2)$, while the offer $1/2$ is always accepted since $A(1/2;\hat{\omega}^{R,U}) > 0$. Since $\hat{\tau} \in [0, 1/2]$, the best accepted offer is the maximizer of W on $[\hat{\tau}, 1/2]$, namely $\max\{s_D^*(\omega^S), \hat{\tau}\}$. This accepted offer weakly dominates rejection: if the maximum is s_D^*, it is the best accepted offer; if the maximum is $\hat{\tau}$, then $\hat{\tau} \in [s_D^*, 1/2]$ and concavity implies $W(\hat{\tau}) \geq W(1/2) > 0$.

Dividing $\Phi_D(\hat{\tau};\omega^S)$ by $\omega_d^S > 0$ gives the factorization (II.5), so the sign of the marginal value $\Phi_D(\hat{\tau};\omega^S)$ is the sign of $\Gamma(\hat{\tau}) - \kappa^S$. Because Φ_D is strictly decreasing on $[0,1/2]$, if $\hat{\tau} > 0$ we have $s_D^* > \hat{\tau}$ iff $\kappa^S < \Gamma(\hat{\tau})$, $s_D^* < \hat{\tau}$ iff $\kappa^S > \Gamma(\hat{\tau})$, and equality iff $\kappa^S = \Gamma(\hat{\tau})$. If $\kappa^S < \Gamma(\hat{\tau})$ then also $\Phi_D(0;\omega^S) \geq \Phi_D(\hat{\tau};\omega^S) > 0$, so the Dictator optimum is interior and $o^* = s_D^* > 0$. If $\hat{\tau} = 0$ and $\kappa^S > \Gamma(0)$, then $\Phi_D(0;\omega^S) < 0$, so $s_D^* = 0$ and $o^* = 0$. The index $\Gamma(\tau) = 2(1-2\tau)$ is strictly decreasing on $[0,1/2]$.

For item 4: by Proposition II.2, $\hat{\tau} = 0$ iff $\hat{\omega}_e^R \geq \hat{\omega}_d^R/2$, in which case $\Gamma(\hat{\tau}) = \Gamma(0) = 2$. Otherwise $\hat{\tau}$ is the displayed root and

$$1 - 2\hat{\tau} = \frac{\sqrt{\hat{D}} - \hat{\omega}_o^R}{2\hat{\omega}_d^R}, \qquad \text{so} \qquad \Gamma(\hat{\tau}) = \frac{\sqrt{\hat{D}} - \hat{\omega}_o^R}{\hat{\omega}_d^R}.$$

Since $\sqrt{\hat{D}} \gtrless \hat{\omega}_o^R + 2\hat{\omega}_d^R$ iff $\hat{\omega}_e^R \gtrless \hat{\omega}_d^R/2$, the two expressions combine into the stated minimum, attained by the first argument exactly in the believed-soft case. The case list then instantiates items 1–3; in the believed-soft case $\Gamma(\hat{\tau}) = 2$, and $\kappa^S \geq 2$ merges the strategic corner ($\kappa^S > 2$) with the kink at zero ($\kappa^S = 2$), where the offer is zero either way. □

Proof of Proposition II.4

The offer is the pointwise maximum of two continuous functions, hence continuous. Each branch is monotone in each parameter with the direction claimed: s_D^* is nonincreasing in ω_o^S, nondecreasing in ω_d^S, constant in ω_e^S and in beliefs (Proposition II.1); $\hat{\tau}$ is nonincreasing in $\hat{\omega}_o^R$ and $\hat{\omega}_e^R$, nondecreasing in $\hat{\omega}_d^R$, constant in own weights (Proposition II.2). A pointwise maximum of functions that are all monotone in a given direction is monotone in that direction, including at the kink, where the one-sided derivatives are those of the two branches. This proves global monotonicity; wherever a derivative of o^* exists it therefore carries the stated weak sign, and at kink points both one-sided derivatives do.

Strictness: if $\kappa^S < \Gamma(\hat{\tau})$ then $s_D^* > \hat{\tau} \geq 0$ by Proposition II.3, so the active branch is the interior Dictator transfer, which is strictly monotone in ω_o^S and ω_d^S; the belief branch is inactive, so small belief changes do not move the offer. If $\kappa^S > \Gamma(\hat{\tau})$ and $\hat{\tau} > 0$ the active branch is the interior believed minimum acceptable offer, which is strictly monotone in all three believed weights, while the own-weight branch is inactive. At $\kappa^S = \Gamma(\hat{\tau})$ the two branches coincide and one-sided derivatives apply.

Regime switching: the regime gap $\Gamma(\hat{\tau}) - \kappa^S$ is strictly decreasing in ω_o^S and strictly increasing in ω_d^S, through $\kappa^S = \omega_o^S/\omega_d^S$ only, and constant in ω_e^S. Since Γ is strictly decreasing, the gap inherits the opposite of the monotonicity of $\hat{\tau}$ in the believed weights (Proposition II.2), which is strict whenever $\hat{\tau}$ is interior. Monotonicity of the gap in each parameter implies the single-switch property, and the description of the paths follows by combining the branch formulas with the switch direction. □

Proof of Corollary II.1

Under full projection $\hat{\omega}^{R,U} = \omega^S$, so the believed minimum acceptable offer in Proposition II.3 is $\tau^*(\omega^S)$, and the offer formula and regime characterization follow directly. For the comparative statics: under projection both branches are monotone in the *same* direction in ω_o^S and ω_d^S — the Dictator transfer is strictly decreasing in ω_o^S and strictly increasing in ω_d^S whenever interior, and the projected minimum acceptable offer is strictly decreasing in ω_o^S and strictly increasing in ω_d^S whenever interior (Propositions II.1 and II.2). If $o^P > 0$ the active branch (or, at the kink, both branches) is interior and strictly monotone, so the maximum is strictly monotone. For ω_e^S, the Dictator branch is constant and the projected acceptance branch is nonincreasing, so the maximum is nonincreasing, strictly when the acceptance branch is active and interior. $\square$

Proof of Proposition II.5

The return objective is strictly concave in r ($\Phi_{R,r} = -4\omega_d^R < 0$), and $\Phi_R(ms, s; \omega^R) = -\omega_o^R + 2\omega_d^R((1-m)s-1) < 0$, so the upper corner never binds. The first-order condition $-\omega_o^R + 2\omega_d^R\big((m+1)s-1-2r\big) = 0$ solves to the displayed closed form; the return is positive exactly when $s > \bar{s}_R = (1+2k)/(m+1)$, which is interior iff $k < m/2$, and at a positive return $g_R(r^*, s) = (m+1)s-1-2r^* = 2k$, i.e. $y_R - y_S = 2k$. Differentiating the closed form gives the slope $(m+1)/2$ and the local comparative statics, and the cutoff statics follow from $\bar{s}_R = (1+2k)/(m+1)$. For the global statement, evaluate Φ_R at the current optimum after the weight change: increasing ω_o^R lowers Φ_R pointwise, so by strict concavity the maximizer weakly falls and the corner $r^* = 0$ persists; increasing ω_d^R raises Φ_R at an interior optimum (where $g_R = 2k > 0$) and at a corner with $g_R(0, s) \geq 0$, so the maximizer weakly rises, while a corner with $g_R(0, s) < 0$ has $\Phi_R(0, s) < 0$ for all weights and persists; ω_e^R does not enter (II.8). $\square$

Proof of Lemma II.1

(1) On the cautious region $\hat{r} = 0$, so

$$T(s) = T_0(s) \equiv \omega_o^S(1-s) + \omega_e^S\big(1+(m-1)s\big) - \frac{\omega_d^S}{2}\big(1-(m+1)s\big)^2,$$

a quadratic with $T_0'' = -\omega_d^S(m+1)^2 < 0$ and first-order condition $-\omega_o^S + \omega_e^S(m-1) + \omega_d^S(m+1)\big(1-(m+1)s\big) = 0$, whose root is $\tilde{s}_C$; strict concavity makes the clipped candidate unique.

(2) At $s = \hat{s}_R$, $\hat{r} = 0$ and $\hat{d}(\hat{s}_R) = 1-(m+1)\hat{s}_R = -2\hat{k} < 0$. Subtracting the one-sided versions of (II.10), with $\hat{r}_s(\hat{s}_R^+) = \frac{m+1}{2}$,

$$T'_+(\hat{s}_R) - T'_-(\hat{s}_R) = \hat{r}_s(\hat{s}_R^+)\big[\omega_o^S - 2\omega_d^S\hat{d}(\hat{s}_R)\big] = \frac{m+1}{2}\big[\omega_o^S + 4\omega_d^S\hat{k}\big] > 0.$$

(3) For $s > \hat{s}_R$, $\hat{r}(s) = \frac{(m+1)s-1}{2} - \hat{k}$, so

$$\hat{y}_S(s) = \frac{1+(m-1)s}{2} - \hat{k}, \qquad \hat{y}_R(s) = \frac{1+(m-1)s}{2} + \hat{k}, \qquad \hat{d}(s) = -2\hat{k},$$

and substituting into (II.9),

$$T(s) = \omega_o^S\left(\frac{1+(m-1)s}{2} - \hat{k}\right) + \omega_e^S\big(1+(m-1)s\big) - 2\omega_d^S\hat{k}^2,$$

affine and strictly increasing with slope $(m-1)\big(\omega_o^S/2 + \omega_e^S\big) > 0$; its unique maximizer on $[\hat{s}_R, 1]$ is $s = 1$, and the believed return and payoffs at $s = 1$ follow by substitution.

(4) By (2)–(3), $T'_+(\hat{s}_R) > 0$, so the cutoff is dominated by sends just above it and is never optimal; non-concavity on $[0, 1]$ is immediate from (2) since concave functions have nonincreasing derivatives. Combining (1) and (3), the global optimum is either the cautious candidate s_C or full sending. $\square$

Proof of Proposition II.6

(1) *Returns.* The believed return $\hat{r}(s) = \max\{0, \frac{(m+1)s-1}{2} - \hat{k}\}$ is nonincreasing in $\hat{k}$, and pessimism raises $\hat{k} = \hat{\omega}_o^R/(4\hat{\omega}_d^R)$, so $r^*(s;\hat{\omega}') \leq r^*(s;\hat{\omega})$ for every s.

Values. Fix s with $r^* \equiv r^*(s;\hat{\omega}) > 0$ and consider the sender's utility as a function of the return level r,

$$\psi(r) = \omega_o^S(1 - s + r) + \omega_e^S q(s) - \frac{\omega_d^S}{2}\big(1 - (m+1)s + 2r\big)^2, \qquad \psi'(r) = \omega_o^S - 2\omega_d^S \hat{d}(s,r),$$

where $\hat{d}(s,r) = 1 - (m+1)s + 2r$. Receiver optimality gives $\hat{d}(s,r^*) = -2\hat{k} < 0$, and $\hat{d}(s,\cdot)$ is increasing, so $\hat{d}(s,r) < 0$ for all $r \leq r^*$; hence $\psi' > 0$ on $[0,r^*]$. Since $\hat{r}(s;\hat{\omega}') \in [0,r^*]$, we get $T(s;\hat{\omega}') = \psi(\hat{r}(s;\hat{\omega}')) \leq \psi(\hat{r}(s;\hat{\omega})) = T(s;\hat{\omega})$. Where $\hat{r}(s;\hat{\omega}) = 0$ both returns are zero and the values coincide; in particular this holds on the cautious region, so the cautious candidate and its value are unaffected.

(2) Suppose every optimal send under $\hat{\omega}$ is cautious and let s_0 be one of them, so $T(s;\hat{\omega}) < T(s_0;\hat{\omega})$ for every s with $\hat{r}(s;\hat{\omega}) > 0$. Take any s with $\hat{r}(s;\hat{\omega}') > 0$. By (1), $\hat{r}(s;\hat{\omega}) \geq \hat{r}(s;\hat{\omega}') > 0$, hence

$$T(s;\hat{\omega}') \leq T(s;\hat{\omega}) < T(s_0;\hat{\omega}) = T(s_0;\hat{\omega}'),$$

the last equality because $\hat{r}(s_0;\hat{\omega}) = 0$ implies $\hat{r}(s_0;\hat{\omega}') = 0$. So every trusting send is strictly suboptimal under $\hat{\omega}'$.

(3) Immediate from pointwise dominance. Independence of $\hat{\omega}_e^R$: the believed return function does not depend on it, so neither does T. □

Proof of Proposition II.7

(1) At an interior cautious optimum, $s_C = \tilde{s}_C$, and differentiating the closed form in Lemma II.1(1) gives

$$\frac{\partial s_C}{\partial \omega_o^S} = -\frac{1}{\omega_d^S(m+1)^2} < 0, \qquad \frac{\partial s_C}{\partial \omega_e^S} = \frac{m-1}{\omega_d^S(m+1)^2} > 0, \qquad \frac{\partial s_C}{\partial \omega_d^S} = \frac{\omega_o^S - \omega_e^S(m-1)}{(\omega_d^S)^2(m+1)^2},$$

whose sign is that of $\omega_o^S - \omega_e^S(m-1)$, i.e. of $1-(m+1)s_C$ (from the first-order condition). Beliefs enter T only through $\hat{r}$, which is locally zero and locally invariant on the interior of the cautious region, so s_C is locally independent of beliefs.

(2) Each branch value is a maximum of functions affine in ω^S over a compact set, hence convex and continuous in ω^S. By Lemma II.1(3) the trusting branch value is simply $T(1)$, affine in ω^S with coefficients $\big(\frac{m}{2} - \hat{k},\, m,\, -2\hat{k}^2\big)$; the cautious branch value has the coefficients $\big(1 - s_C,\, q(s_C),\, -\frac{1}{2}(1 - (m+1)s_C)^2\big)$ at its maximizer s_C, which is unique by Lemma II.1(1) (envelope theorem). Subtracting gives the displayed derivatives. For claim (i): $1 > \hat{s}_R \geq s_C$, so $\partial G/\partial \omega_e^S = (m-1)(1-s_C) > 0$. Note also that wherever $G \leq 0$ the cautious candidate is interior to the cautious region: if $s_C = \hat{s}_R$, continuity of T at the cutoff and the strict increase of T beyond it (Lemma II.1(3)) imply $G > 0$. A continuous function of ω_e^S that is strictly increasing wherever it is nonpositive crosses zero at most once from below, which gives the threshold. □

Proof of Proposition II.8

(1) If $\hat{k} \geq m/2$ then $\hat{s}_R \geq 1$ and $\hat{r} \equiv 0$ on $[0,1]$, so the sender's objective is the strictly concave cautious quadratic T_0 of Lemma II.1(1), and the optimum is the projection of $\tilde{s}_C$ on $[0,1]$.

(2) $T(1)$ is strictly decreasing in $\hat{k}$ (derivative $-\omega_o^S - 4\omega_d^S\hat{k} < 0$). $T(s_C)$ is nondecreasing in $\hat{k}$: when $s_C < \hat{s}_R$ it is locally constant in $\hat{k}$, and when $s_C = \hat{s}_R$ it equals $T_0(\hat{s}_R)$ with $\hat{s}_R$ increasing in $\hat{k}$ and T_0 increasing on $[0, \tilde{s}_C] \supseteq [0, \hat{s}_R]$. Hence G is continuous and strictly decreasing in $\hat{k}$ and crosses zero at most once; set $\bar{k} = \sup\{\hat{k} \in [0, m/2) : G(\hat{k}) \geq 0\}$, with $\bar{k} = -\infty$ if the set is empty. As $\hat{k} \to m/2$, $\hat{s}_R \to 1$ and $T(1) \to T_0(1) \leq T_0(s_C)$, so $G \leq 0$ in the limit and $\bar{k} \leq m/2$. Implicit differentiation of $G(\bar{k}) = 0$, with $G_{\hat{k}} < 0$, transfers the signs of Proposition II.7(2) to $\bar{k}$. For the multiplier: $\partial T(1)/\partial m = \omega_o^S/2 + \omega_e^S$ (holding $\hat{k}$ fixed), and by the envelope theorem $\partial T_0(s_C)/\partial m = s_C\big[\omega_e^S + \omega_d^S(1-(m+1)s_C)\big]$. If $s_C = 0$ then $\partial G/\partial m = \omega_o^S/2 + \omega_e^S > 0$. If $s_C = \tilde{s}_C$ is interior, the cautious first-order condition gives $\omega_d^S\big(1-(m+1)s_C\big) = \frac{\omega_o^S - \omega_e^S(m-1)}{m+1}$, so

$$\frac{\partial G}{\partial m} = \omega_o^S\Big(\frac{1}{2} - \frac{s_C}{m+1}\Big) + \omega_e^S\Big(1 - \frac{2s_C}{m+1}\Big) > 0,$$

using $s_C \leq 1 < \frac{m+1}{2}$. Since caution implies $s_C = \max\{0, \tilde{s}_C\}$ by item 3, this covers the whole no-trust region, so G single-crosses zero from below in m.

(3) If the sender is cautious then $\tilde{s}_C < \hat{s}_R$: otherwise $s_C = \hat{s}_R$ and, since T is continuous at $\hat{s}_R$ and strictly increasing beyond it, $T(1) > T(\hat{s}_R) = T(s_C)$, contradicting caution. Hence $s_C = \max\{0, \tilde{s}_C\}$ strictly below the cutoff. □

Proof of Proposition II.9

Let T_0 be the belief-free cautious objective of Lemma II.1(1) and $\sigma_C = \min\{\max\{\tilde{s}_C, 0\}, 1\}$ its maximizer on $[0,1]$. Whenever the sender does not trust, his send is σ_C: if the trusting region is empty this is immediate; otherwise caution forces $s_C < \hat{s}_R$ (Proposition II.8(3)), and then $s_C = \min\{\sigma_C, \hat{s}_R\} = \sigma_C$. In particular the send under caution does not depend on beliefs. Whenever the sender trusts, his send is 1 (Lemma II.1(3)). So for every belief,

$$s_T^* \in \{\sigma_C,\, 1\}, \qquad \sigma_C \leq 1,$$

with $s_T^* = 1$ exactly when the sender trusts, i.e. when $\hat{k} \leq \bar{k}$ (Proposition II.8).

Beliefs. Fix ω^S and make the belief more pessimistic ($\hat{\omega}_o^R$ weakly up, $\hat{\omega}_d^R$ weakly down, $\hat{\omega}_e^R$ arbitrary). By Proposition II.6(2) the trust decision can only switch from trusting to caution, so the send moves (if at all) from 1 down to σ_C: it is nonincreasing. Coordinate-wise monotonicity in $\hat{\omega}_o^R$ and $\hat{\omega}_d^R$, and constancy in $\hat{\omega}_e^R$, are special cases, and the step function is as displayed.

Efficiency. Within the cautious regime $s_T^* = \sigma_C$ is nondecreasing in ω_e^S: $\partial\Phi_T/\partial\omega_e^S = m-1 > 0$ pointwise, and a pointwise upward shift of the marginal payoff moves the maximizer of the strictly concave T_0 weakly up, including at the corners. Within the trust regime $s_T^* = 1$ is constant. By Proposition II.7(2), G is continuous and strictly increasing in ω_e^S wherever $G \leq 0$, so as ω_e^S increases the regime switches at most once, from caution to trust; at the switch the send jumps from $\sigma_C < \hat{s}_R < 1$ up to 1. A function that is nondecreasing within regimes and jumps upward at its unique switch is globally nondecreasing.

Failure for the remaining parameters. For ω_d^S and m the within-regime effect on the cautious send is not uniformly signed (Proposition II.7(1)). For ω_o^S, take $\omega_e^S = 0$, $\omega_d^S = 1$, $m > 2$, and $0 < \hat{k} < \frac{m}{2} - 1$. As $\omega_o^S \downarrow 0$, $G \to -2\hat{k}^2 < 0$, so the sender is cautious with send near $1/(m+1)$, decreasing in ω_o^S. For $\omega_o^S \geq m+1$ the cautious candidate is $s_C = 0$ and $G = \omega_o^S\big(\frac{m}{2} - \hat{k} - 1\big) + \frac{1}{2} - 2\hat{k}^2$, which is strictly increasing and eventually positive, so the sender eventually trusts fully. The send therefore falls with ω_o^S over the cautious range and then jumps to 1: it is not globally monotone. □

Proof of Corollary II.2

(1) By Lemma II.1(3) the trusting value is $V_T = T(1) = \omega_o^S(\frac{m}{2} - \hat{k}) + \omega_e^S m - 2\omega_d^S \hat{k}^2$, linear in ω_o^S, while the cautious value $V_C = \max_{s\in[0,\hat{s}_R]} T(s)$ is a maximum of functions linear in ω_o^S, hence convex. Thus $G = V_T - V_C$ is concave in ω_o^S, its superlevel set $\{G \geq 0\}$ is an interval, and the regime switches at most twice as ω_o^S increases: into trust, then out.

(2) By the envelope theorem, at points of differentiability $\partial G/\partial\omega_o^S = (\frac{m}{2} - \hat{k}) - (1 - s_C)$. As $\omega_o^S \to \infty$, $\tilde{s}_C \to -\infty$, so $s_C \to 0$ and $\partial G/\partial\omega_o^S \to (\frac{m}{2} - \hat{k}) - 1$. If $\hat{k} < \frac{m}{2} - 1$ this limit is positive and $G \to +\infty$: the sender trusts for all large ω_o^S. If $\hat{k} > \frac{m}{2} - 1$ it is negative and $G \to -\infty$: the sender is cautious for all large ω_o^S, with send $s_C \to 0$. □

Proof of Proposition II.10

By homogeneity of degree one of utility in the weight vector (and degree zero of $\hat{k}$, $\hat{s}_R$, and $\tilde{s}_C$), the trust decision depends only on (κ^S, e^S, m); normalize $\omega_d^S = 1$, so $\hat{k} = \kappa^S/4$ and $\hat{s}_R = \frac{1+\kappa^S/2}{m+1}$. Write $H = \{(\kappa^S, e^S) : \kappa^S > 0,\ e^S \geq 0,\ e^S(m-1) \leq \kappa^S \frac{m+3}{2}\}$ for the set on which the cautious candidate is not constrained by the cutoff (the inequality is $\tilde{s}_C \leq \hat{s}_R$, rearranged).

Off H the sender trusts. If $\tilde{s}_C > \hat{s}_R$ then $s_C = \hat{s}_R$ and, exactly as in the proof of Proposition II.8(3), continuity of T at the cutoff and strict increase beyond it give $T(1) > T(s_C)$, so $G > 0$.

On H, G is concave in (κ^S, e^S) jointly. On H the cautious value equals the unconstrained value $V_C(\kappa^S, e^S) = \max_{s\in[0,1]} T_0(s; \kappa^S, e^S)$, a maximum of functions affine in (κ^S, e^S), hence convex; and $T(1) = \frac{m}{2}\kappa^S - \frac{3}{8}(\kappa^S)^2 + e^S m$ is a concave quadratic (substituting $\hat{k} = \kappa^S/4$). So $G = T(1) - V_C$ is concave on the convex set H, continuous, with $G(0, 0) = 0$.

Threshold in κ^S and monotonicity in e^S. On H, $\partial G/\partial e^S = (m-1)(1-s_C) > 0$ wherever $s_C < 1$, which holds throughout the no-trust region. Fix e^S and increase κ^S: the set of κ^S with $(\kappa^S, e^S) \in H$ is a half-line $\{\kappa^S \geq \kappa_a(e^S)\}$, G is concave along it, and $G > 0$ at its left endpoint — either $\kappa_a(e^S) > 0$, which lies on the boundary of H where $\tilde{s}_C = \hat{s}_R$ and the trusting candidate strictly dominates by the first step, or $\kappa_a(e^S) = 0$ (the case $e^S = 0$), where $G \to 0$ as $\kappa^S \downarrow 0$ with strictly positive right derivative $\frac{m(m-1)}{2(m+1)}$. A concave function positive at the left endpoint of a half-line is nonnegative exactly on an interval attached to that endpoint; joined with the all-trust region off H, the trust region at fixed e^S is an interval $[0, \bar{\kappa}(e^S, m)]$ with $\bar{\kappa} > 0$. That $\bar{\kappa}$ is nondecreasing in e^S follows from $\partial G/\partial e^S > 0$ on H together with the fact that the complement of H is an upper set in e^S at fixed κ^S. That $\bar{\kappa} \leq 2m$ follows because $\kappa^S \geq 2m$ means $\hat{k} \geq m/2$ and full trust is infeasible; indeed as $\kappa^S \to 2m$, $T(1) \to T_0(1) \leq V_C$, so $G \leq 0$ there.

Monotonicity in the raw weights. Increasing ω_o^S raises κ^S leaving e^S fixed, so the trust indicator can only switch from 1 to 0. Increasing ω_e^S raises e^S leaving κ^S fixed, so by the previous paragraph the indicator can only switch from 0 to 1. Increasing ω_d^S scales (κ^S, e^S) toward the origin along a ray. Rays through the origin either lie entirely in H or entirely outside it, because the constraint defining H is homogeneous. Outside H, trust always holds. Inside, $t \mapsto G(t\kappa_0, te_0)$ is concave with value 0 at $t = 0$ and strictly positive right derivative $\frac{m(m-1)}{2(m+1)}\kappa_0 + \frac{m(m-1)}{m+1}e_0 > 0$, so $\{t \geq 0 : G \geq 0\}$ is an interval $[0, \bar{t}]$; lowering t (raising ω_d^S) can only switch the indicator from 0 to 1.

Monotonicity in m. Under projection $\hat{k}$ does not depend on m, so the computation in the proof of Proposition II.8(4) applies verbatim: $\partial G/\partial m > 0$ throughout the no-trust region, giving single crossing.

Item 3. In the cautious regime $G < 0$ strictly, so small parameter changes keep the sender cautious, and $\tilde{s}_C < \hat{s}_R$: a cautious candidate clipped at the cutoff would be strictly dominated by full trust, as shown in the first step. On a neighborhood of $\tilde{s}_C$ the believed return is identically zero, and it remains so after small changes in ω^S because the projected cutoff $\hat{s}_R = \frac{1+\kappa^S/2}{m+1}$ moves continuously. Only the preference channel therefore operates locally, and the signs follow by differentiating the closed form of $\tilde{s}_C$, as in Proposition II.7(1).

Item 4. When $0 \le \tilde{s}_C \le \hat{s}_R$, the vertex value of the cautious quadratic is

$$T_0(\tilde{s}_C) = \frac{m}{m+1}\kappa^S + \frac{2m}{m+1}e^S + \frac{\big(e^S(m-1)-\kappa^S\big)^2}{2(m+1)^2},$$

and subtracting from $T(1)$ gives $\tilde{G}$. At $e^S = 0$,

$$\tilde{G}(\kappa^S, 0) = \kappa^S\left[\frac{m(m-1)}{2(m+1)} - \left(\frac{3}{8} + \frac{1}{2(m+1)^2}\right)\kappa^S\right],$$

whose positive root is the displayed $\bar{\kappa}(0,m)$.

Item 5. Within the trust regime the send is constant at 1; within the cautious regime it equals $\max\{0, \tilde{s}_C\}$, decreasing in ω_o^S and increasing in ω_e^S; and by item 2 each of these parameters switches the regime at most once — from trust to cautious as ω_o^S increases, with the send dropping from 1 to $\max\{0, \tilde{s}_C\} < \hat{s}_R < 1$, and from cautious to trust as ω_e^S increases, with the send jumping up to 1. Monotone within regimes with a single like-signed jump implies global monotonicity. For ω_d^S and m the cautious-send effect is not uniformly signed (item 3), so no global sign is possible. □

Proof of Proposition II.11

The return objective is strictly concave in r, with marginal value $\Phi(r,s) = -\omega_o^R + \omega_d^R(s-r)$ and $\partial\Phi/\partial r = -\omega_d^R < 0$; at $r = s$, $\Phi = -\omega_o^R < 0$, so $r^* < s \le ms$: the receiver never fully repays and the upper corner never binds. The first-order condition solves to the displayed closed form $r^* = \max\{0, s-c\}$: the return is positive exactly above the cutoff $\bar{s}_I = c$, interior exactly when $c < 1$, and the marginal return is one. Differentiating the closed form gives the local comparative statics and the cutoff statics. The global weak versions follow because Φ is pointwise decreasing in ω_o^R, increasing in ω_d^R at any optimum with $s - r \ge 0$ (the corner at $s = 0$ persists), and free of ω_e^R, and a pointwise shift of a strictly concave objective's marginal value moves the maximizer weakly in the same direction. □

Proof of Lemma II.2

(1) On the cautious region $\hat{r} = 0$, so $T = T_0$, the belief-free cautious objective displayed in the text, strictly concave with unconstrained maximizer $\tilde{s}_C$ (Lemma II.1(1)).

(2) For $s > \hat{c}$, $\hat{r}(s) = s - \hat{c}$ gives the believed payoffs displayed in the text; substituting into (II.12),

$$T(s) = \omega_o^S(1-\hat{c}) + \omega_e^S\big(1+(m-1)s\big) - \frac{\omega_d^S}{2}\big(1-2\hat{c}-(m-1)s\big)^2,$$

a quadratic with $T'' = -\omega_d^S(m-1)^2 < 0$ and maximizer $\tilde{s}_I$.

(3) As in the proof of Lemma II.1(2), the one-sided derivatives at the cutoff differ by $\hat{r}_s(\hat{c}^+)\big[\omega_o^S - 2\omega_d^S\hat{d}(\hat{c})\big] = \omega_o^S - 2\omega_d^S\big(1-(m+1)\hat{c}\big)$ — but $\hat{d}(\hat{c}) = 1-(m+1)\hat{c}$ is no longer signed, because the norm-based cutoff condition does not force the receiver ahead of the sender. When the jump is negative, T' is decreasing on each branch and falls at the kink, so T is strictly concave on $[0,1]$.

(4) A convex kink cannot be a local maximum, by the argument in the proof of Lemma II.1(4). If the kink is concave then $d_0 \equiv 1-(m+1)\hat{c} > 0$ and $T'_+(\hat{c}) = (m-1)\big[\omega_e^S + \omega_d^S d_0\big] > 0$, so the

cutoff is dominated from the right in the concave case too. By strict concavity of each branch, the maximum over the cautious region is s_C and the maximum over the trusting region is $\min\{\tilde{s}_I, 1\}$ when $\tilde{s}_I > \hat{c}$; when $\tilde{s}_I \le \hat{c}$ the trusting branch is nonincreasing, its supremum is the cutoff value $T_0(\hat{c}) \le T_0(s_C)$, and every optimal send is cautious. □

Proof of Proposition II.12

(1) The two-candidate structure is Lemma II.2(4), and under correct beliefs the return met by each candidate is given by Proposition II.11: zero at $s_C \le \hat{c} = \bar{s}_I$, and $s_T^* - c \in (0, m s_T^*)$ at a trusting send. For openness: at $m = 3$, $\omega^S = (1, 0.5, 1)$, $\hat{c} = 0.3$: $\tilde{s}_I = 0.45 \in (0.3, 1)$, $s_C = \tilde{s}_C = 1/4$, and $G = 1.525 - 1.5 > 0$; all quantities are continuous in $(\omega^S, \hat{c}, m)$ and all inequalities strict, so the configuration survives on an open neighborhood, and the equilibrium return is $s_T^* - c = 0.15$.

(2) Differentiate $\tilde{s}_I$.

(3) Wherever $G \le 0$ the cautious candidate is unclipped, $s_C = \max\{0, \tilde{s}_C\} < \hat{c}$: if $s_C = \hat{c}$ then $T_0'(\hat{c}) \ge 0$, and since the cutoff is dominated from the right (Lemma II.2(3)–(4)), T strictly increases past it and $G > 0$, a contradiction. *Own weight:* on the whole trusting branch $\hat{y}_S \equiv 1 - \hat{c}$, including at the clipped cutoff where $\hat{r} = 0$, so by the envelope theorem $\partial G/\partial \omega_o^S = (1 - \hat{c}) - (1 - s_C) = s_C - \hat{c} < 0$. *Efficiency:* $\partial G/\partial \omega_e^S = q(s_+) - q(s_C) > 0$ as before. *Beliefs:* at an interior trusting optimum the first-order condition gives $\hat{d}(\tilde{s}_I) = -\omega_e^S/\omega_d^S \le 0$, so $\partial T(\tilde{s}_I)/\partial \hat{c} = -\omega_o^S + 2\omega_d^S \hat{d}(\tilde{s}_I) = -\omega_o^S - 2\omega_e^S < 0$; in the clipped case $\partial T_0(\hat{c})/\partial \hat{c} = T_0'(\hat{c}) < 0$; and at the corner $s_+ = 1$, $\partial T(1)/\partial \hat{c} = -\omega_o^S + 2\omega_d^S\big(2 - m - 2\hat{c}\big)$, which is negative unless $2 - m - 2\hat{c} > \kappa^S/2 > 0$ — a case incompatible with $G \le 0$, because $\hat{c} < \frac{2-m}{2} - \frac{\kappa^S}{4}$ implies

$$\tilde{s}_C - \hat{c} \;\ge\; \frac{1}{m+1} - \frac{\kappa^S}{(m+1)^2} - \hat{c} \;>\; \frac{m(m-1)}{2(m+1)} + \frac{\kappa^S(m-1)(m+3)}{4(m+1)^2} \;>\; 0,$$

so the cautious candidate would be clipped at the cutoff and the sender would trust. Hence G is strictly decreasing in ω_o^S and $\hat{c}$ and strictly increasing in ω_e^S wherever $G \le 0$, which yields the thresholds.

(4) Within the cautious regime the send $\max\{0, \tilde{s}_C\}$ falls in ω_o^S, rises in ω_e^S, and is constant in $\hat{c}$; within the trust regime it is $\min\{\tilde{s}_I, 1\}$, constant in ω_o^S, nondecreasing in ω_e^S, and nonincreasing in $\hat{c}$. By item 4 each of the three parameters switches the regime at most once — out of trust as ω_o^S or $\hat{c}$ rises, into trust as ω_e^S rises — and at the switch the send jumps in the like direction, between a cautious send below $\hat{c}$ and a trusting send strictly above it. Monotone within regimes with a single like-signed jump implies global monotonicity. For ω_d^S the cautious send strictly rises when $s_C < 1/(m+1)$ while the trusting send strictly falls when $\omega_e^S > 0$; for m the trusting send strictly falls while the cautious send can strictly rise (e.g. $\omega_o^S = 0$, ω_e^S/ω_d^S large, $m < 3$); so no global statement holds for either. □

Proof of Proposition II.13

(1) If $\kappa^S \ge 1$ the believed cutoff $\hat{c} = \kappa^S$ leaves the unit interval, $\hat{r} \equiv 0$, and $T = T_0$. For $\kappa^S < 1$ the threshold follows from item 3: wherever $G \le 0$, G is strictly decreasing in ω_o^S (hence in κ^S at fixed ω_d^S) and strictly increasing in ω_e^S, so each parameter crosses the trust boundary at most once.

(2) Substitute $\hat{c} = \kappa^S$ into Lemma II.2(2) and differentiate. Openness: at $m = 3$, $\omega^S = (0.3, 0.5, 1)$: $\tilde{s}_I = 0.45 \in (0.3, 1)$, $s_C = \tilde{s}_C = 0.29375$, and $G = 1.035 - 0.990 > 0$, with all inequalities strict.

(3) *Own weight.* Wherever $G \le 0$ the cautious candidate is unclipped, $s_C < \hat{c} = \kappa^S$ (the argument in the proof of Proposition II.12(3)). *Interior*: $dG/d\omega_o^S = \big[(1 - \kappa^S) - (1 - s_C)\big] - \kappa^S -$

$2e^S = s_C - 2\kappa^S - 2e^S < 0$. *Corner*: $dG/d\omega_o^S = (s_C - \kappa^S) - \kappa^S + 2(2 - m - 2\kappa^S)$; for $m \geq 2$ both pieces are negative, and for $m < 2$ positivity would require $2 - m - 2\kappa^S > \kappa^S/2$, which is incompatible with caution by the argument in the proof of Proposition II.12(3) with $\hat{c} = \kappa^S$. *Clipped*: $dG/d\omega_o^S = (s_C - \kappa^S) + T_0'(\kappa^S)/\omega_d^S < 0$. Within regimes the send is nonincreasing ($\tilde{s}_C$ falls, $d\tilde{s}_I/d\omega_o^S = -2/((m-1)\omega_d^S) < 0$, the corner is constant) and the single switch jumps down: global monotonicity follows.

Efficiency. The belief channel is null (Proposition II.11), so the free-belief argument in the proof of Proposition II.12 applies verbatim.

Failures. For ω_d^S at $m = 3$ the send strictly rises at $(\omega_o^S, \omega_e^S, \omega_d^S) = (1.70, 0.67, 2.16)$ and strictly falls at $(0.05, 1.18, 1.28)$ (verified directly); for m the within-caution effect is not uniformly signed, as in Proposition II.10(5). □

II.B Robustness: Concave Earnings Utility

Throughout the paper, utility is linear in own and in total earnings. This section replaces the two earnings terms with a concave index and records which results survive. Let $u : [0, m] \to \mathbb{R}$ be twice continuously differentiable, with finite endpoint derivatives, $u(0) = 0$, $u' > 0$, and $u'' \leq 0$ — call such a u *admissible* — and let

$$U^p = \omega_o^p\, u(y_p) + \omega_e^p\, u(y_p + y_{-p}) - \frac{\omega_d^p}{2}(y_p - y_{-p})^2,$$

the disparity term unchanged; the model of the main text is the case $u(z) = z$. Beliefs are point beliefs as before, with believed returns recomputed under u. Three propositions collect what survives; everything else — the closed forms, the one-parameter belief statistics $\hat{k}$ and $\hat{c}$, the threshold characterizations, and the remaining global signs — is linear.

Proposition II.14 (Concave Problems Are Robust)**.** *Let u be admissible.*

1. (Dictator.) *The transfer is unique; it is zero iff $2\omega_d^S \leq \omega_o^S u'(1)$ and interior — in $(0, 1/2)$ — otherwise; it is globally nonincreasing in ω_o^S, constant in ω_e^S, and nondecreasing in ω_d^S.*

2. (Ultimatum.) *The minimum acceptable offer is unique on $[0, 1/2]$; it is zero iff $\omega_e^R u(1) \geq \omega_d^R/2$ and interior otherwise, globally nonincreasing in ω_o^R and ω_e^R and nondecreasing in ω_d^R. The offer is still $o^* = \max\{s^*, \hat{\tau}\}$, the regime is still governed by $\kappa^S \lesseqgtr \Gamma(\hat{\tau})$ with the strictly decreasing toughness index $\Gamma(\tau) = 2(1 - 2\tau)/u'(1 - \tau)$, and the global comparative statics of Proposition II.4 hold verbatim.*

3. (Trust return, gap benchmark.) *The return is unique, zero at or below a cutoff and interior above it, with marginal return $r_s^* = \frac{m\lambda + 2\omega_d^R(m+1)}{\lambda + 4\omega_d^R} > 1$, where $\lambda = \omega_o^R\big[-u''(ms - r^*)\big] \geq 0$; at any positive return the receiver stays ahead of the sender, by $\omega_o^R\, u'(ms - r^*)/(2\omega_d^R)$; the global signs $(-, 0, +)$ hold.*

4. (Trust returns, norm benchmarks.) *For any benchmark β increasing in s with $\beta(0) = 0$ and $\beta(s) \leq ms$ — the repayment norm qualifies — the return is unique and never reaches the benchmark: at a positive return, $\beta(s) - r^* = (\omega_o^R/\omega_d^R)\, u'(ms - r^*)$, a shortfall decreasing in the send. Returns are zero at or below a cutoff and rise above it at rate $r_s^* = \frac{m\lambda + \omega_d^R \beta'}{\lambda + \omega_d^R} \in [\beta', m)$; the global signs $(-, 0, +)$ hold.*

Proposition II.15 (The Trust Send with Concave Utility)**.** *Let u be admissible and let the believed cutoff be interior.*

1. (Efficiency, all benchmarks.) *Under either benchmark, and for any point belief — including the projected one — the optimal send is globally nondecreasing in ω_e^S; in particular, under full projection the trust decision is monotone in ω_e^S.*

2. (All-or-nothing, gap benchmark.) *The two-branch structure of Lemma II.1 survives: the cautious branch is strictly concave, the kink is strictly convex, the trusting branch is strictly increasing, and $s_T^* \in \{s_C, 1\}$ with the cutoff never optimal.*

3. (Beliefs, gap benchmark.) *Propositions II.6 and II.9 hold verbatim: pessimism lowers believed returns and trusting values pointwise, the cautious plan is belief-free, and the send is a step function of beliefs, globally nonincreasing under pessimism.*

4. (Greed interval, gap benchmark.) *Corollary II.2 survives with the limit criterion $u(\hat{r}(1)) \gtrless u(1)$: the trust set in ω_o^S is an interval, and an arbitrarily greedy sender fully trusts if the believed return to full sending exceeds the endowment, $\hat{r}(1) > 1$, and sends nothing if $\hat{r}(1) < 1$.*

Proposition II.16 (The Repayment Norm with Concave Utility)**.** *Let u be admissible and let the believed cutoff $\hat{s}_I$ be interior.*

1. (Shrinking shortfall.) *On the trusting region the believed return is $\hat{r}(s) = s - \hat{c}(s)$, with believed shortfall $\hat{c}(s) = (\hat{\omega}_o^R/\hat{\omega}_d^R)\, u'\big(ms - \hat{r}(s)\big)$, nonincreasing in s and strictly decreasing wherever $u'' < 0$. The believed own payoff $\hat{y}_S(s) = 1 - \hat{c}(s)$ is therefore nondecreasing, and the believed marginal repayment exceeds one wherever $u'' < 0$: the flat fee is a linear-utility property.*

2. (Interior trust.) *Let $m \geq 2$ and suppose $\hat{s}_I < \frac{1}{m+1}$, so the believed gap is positive at the cutoff; at full sending it is negative, $\hat{d}(1) = 2\hat{r}(1) - m < 2 - m \leq 0$, because $\hat{r}(1) < 1$. Then, fixing ω_o^S and ω_e^S, for all sufficiently large ω_d^S every optimal send is trusting and* interior, *with positive believed return; under correct beliefs the equilibrium outcome has an interior send and an interior return.*

Remark II.10 (What exactly linearity buys). The one substantive linear-utility result is the repayment norm's global greed monotonicity. With $u(z) = z$ the believed own payoff on its trusting branch is constant at $1 - \hat{c}$: own-earnings attention is exactly indifferent about the size of a trusting send and unambiguously prefers caution (Proposition II.12). With strictly concave u, item 1 of Proposition II.16 gives greed a weakly positive stake in the trusting margin, and the global sign genuinely fails: with $u(z) = 1 - e^{-3z}$, $m = 3$, believed weights $(\hat{\omega}_o^R, \hat{\omega}_d^R) = (3, 2)$ (believed cutoff ≈ 0.30), $\omega_e^S = 0$, and $\omega_d^S = 2$, the optimal send falls within caution from 0.223 to 0.219 and then jumps *up*, near $\omega_o^S \approx 3.55$, to an interior trusting send ≈ 0.51 that keeps rising in ω_o^S (verified directly): the shrinking fee makes large trusting sends attractive to greedy senders — exactly the mechanism linearity shuts down. The norm models' belief thresholds and global belief monotonicity (Proposition II.12) are likewise established through the linear closed forms; the gap benchmark's belief results, by contrast, are fully robust (Proposition II.15).

Proof of Proposition II.14

(1) *Dictator.* $D_{ss} = \omega_o^S u''(1-s) - 4\omega_d^S < 0$, so D is strictly concave, and $D_s = -\omega_o^S u'(1-s) + 2\omega_d^S(1-2s) < 0$ for $s \geq 1/2$, so no maximizer lies weakly above 1/2; a unique interior root exists exactly when $D_s(0) = 2\omega_d^S - \omega_o^S u'(1) > 0$, and otherwise the corner $s = 0$ is optimal. The implicit-function theorem gives $\partial s^*/\partial \omega_o^S = u'(1-s^*)/D_{ss} < 0$ and $\partial s^*/\partial \omega_d^S = -2(1-2s^*)/D_{ss} > 0$, with ω_e^S absent; the global weak versions follow from the pointwise monotonicity of D_s in the weights, as in the proof of Proposition II.1.

(2) *Ultimatum.* With $A(o) = \omega_o^R u(o) + \omega_e^R u(1) - \frac{\omega_d^R}{2}(1-2o)^2$: for $o < 1/2$, $A_o = \omega_o^R u'(o) + 2\omega_d^R(1-2o) > 0$, and $A(1/2) = \omega_o^R u(1/2) + \omega_e^R u(1) > 0$, so A crosses zero at most once on $[0, 1/2]$ and $\tau^* = 0$ iff $A(0) = \omega_e^R u(1) - \omega_d^R/2 \geq 0$. The implicit-function theorem gives $\partial\tau^*/\partial\omega_o^R = -u(\tau^*)/A_o < 0$, $\partial\tau^*/\partial\omega_e^R = -u(1)/A_o < 0$, and $\partial\tau^*/\partial\omega_d^R = (1-2\tau^*)^2/(2A_o) > 0$, and the global versions follow from the pointwise monotonicity of A, as in the proof of Proposition II.2. For the proposer, the argument in the proof of Proposition II.3 is unchanged with $\Gamma(\tau) = 2(1-2\tau)/u'(1-\tau)$, strictly decreasing because its numerator strictly decreases while its denominator weakly increases ($u'' \leq 0$); the proof of Proposition II.4 uses only the censoring structure and branch monotonicity, so it applies verbatim.

(3) *Gap-benchmark return.* $\Phi_R(r, s) = -\omega_o^R u'(ms - r) + 2\omega_d^R\big((m+1)s - 1 - 2r\big)$ is strictly decreasing in r and strictly negative at $r = ms$, so the return is unique and the upper corner never binds; at an interior return $2\omega_d^R\, g_R(r^*, s) = \omega_o^R u'(ms - r^*) > 0$, i.e. the receiver stays ahead by the stated amount. $F(s) = \Phi_R(0, s)$ is strictly increasing with $F(0) < 0$, giving the cutoff structure. With $\lambda = \omega_o^R[-u''(ms - r^*)] \geq 0$, differentiating the first-order condition gives

$$r_s^* = \frac{m\lambda + 2\omega_d^R(m+1)}{\lambda + 4\omega_d^R}, \qquad r_s^* - 1 = \frac{(m-1)(\lambda + 2\omega_d^R)}{\lambda + 4\omega_d^R} > 0,$$

and the local signs $(-, 0, +)$; the global versions follow from the pointwise-shift argument in the proof of Proposition II.5.

(4) *Norm returns.* $\Phi_\beta(r, s) = -\omega_o^R u'(ms - r) + \omega_d^R\big(\beta(s) - r\big)$ is strictly decreasing in r with $\Phi_\beta(\beta(s), s) = -\omega_o^R u'(ms - \beta(s)) < 0$, so the return is unique with $r^* < \beta(s) \leq ms$, and at a positive return the first-order condition reads $\beta(s) - r^* = (\omega_o^R/\omega_d^R)u'(ms - r^*)$: a shortfall decreasing in s, because $ms - r^*$ strictly increases ($r_s^* < m$) while u' is nonincreasing. $F(s) = \Phi_\beta(0, s) = -\omega_o^R u'(ms) + \omega_d^R \beta(s)$ is strictly increasing with $F(0) < 0$, giving the cutoff. The implicit-function theorem gives $r_s^* = \frac{m\lambda + \omega_d^R \beta'}{\lambda + \omega_d^R}$, with $r_s^* - \beta' = \lambda(m - \beta')/(\lambda + \omega_d^R) \geq 0$ and $m - r_s^* = \omega_d^R(m - \beta')/(\lambda + \omega_d^R) > 0$, and the local signs $(-, 0, +)$; the global versions again follow from pointwise shifts. □

Proof of Proposition II.15

(1) For $s' > s$ and $\omega_e' > \omega_e$,

$$\big[T(s'; \omega_e') - T(s; \omega_e')\big] - \big[T(s'; \omega_e) - T(s; \omega_e)\big] = (\omega_e' - \omega_e)\big[u(q(s')) - u(q(s))\big] > 0:$$

T has strictly increasing differences in (s, ω_e^S) whatever the believed return function, so the set of optimal sends increases in the strong set order and the tie-breaking selection (the largest optimal send) is nondecreasing. Under projection, a change in ω_e^S leaves the projected return unchanged, because no return choice depends on the receiver's efficiency weight (Proposition II.14), so the same argument applies; the projected decision threshold follows because G is continuous and strictly increasing in ω_e^S wherever $G \leq 0$ (envelope theorem, $u(q(1)) > u(q(s_C))$).

(2) On the cautious region $T'' = \omega_o^S u''(1-s) + \omega_e^S(m-1)^2 u''(q) - \omega_d^S(m+1)^2 < 0$. At the cutoff, the one-sided derivatives of T differ by $\hat{r}_s(\hat{s}_R^+)\big[\omega_o^S u'(\hat{y}_S) - 2\omega_d^S \hat{d}\big]$, and the receiver's cutoff condition $\hat{\omega}_o^R u'(m\hat{s}_R) = 2\hat{\omega}_d^R\big((m+1)\hat{s}_R - 1\big)$ forces $\hat{d}(\hat{s}_R) < 0$, so the kink is strictly convex. On the trusting region, with $\lambda = \hat{\omega}_o^R[-u''(ms - \hat{r})] \geq 0$, Proposition II.14(3) gives $\hat{r}_s - 1 > 0$ and $(m+1) - 2\hat{r}_s = -(m-1)\lambda/(\lambda + 4\hat{\omega}_d^R) \leq 0$, while receiver optimality gives $\hat{d} = -\hat{\omega}_o^R u'(ms - \hat{r})/(2\hat{\omega}_d^R) < 0$: in the sender's marginal payoff

$$\Phi_T = \omega_o^S u'(\hat{y}_S)(\hat{r}_s - 1) + \omega_e^S(m-1)u'(q) + \omega_d^S \hat{d}\big((m+1) - 2\hat{r}_s\big),$$

every term is nonnegative and the first strictly positive, so T strictly increases and the trusting candidate is $s = 1$. Hence $s_T^* \in \{s_C, 1\}$ and the cutoff is never optimal, as in Lemma II.1.

(3) Pessimism weakly lowers the return at every send (Proposition II.14(3), applied coordinate-wise), and at any s with a positive believed return the sender's utility is increasing in the return level r on $[0, \hat{r}(s)]$: $\partial T/\partial r = \omega_o^S u'(\hat{y}_S) - 2\omega_d^S \hat{d}(s,r) > 0$, because $\hat{d}(s,r) \leq \hat{d}(s, \hat{r}(s)) < 0$ by receiver optimality. The arguments in the proofs of Propositions II.6 and II.9 then apply verbatim: caution selects the belief-free maximizer of the strictly concave T_0, trusting selects 1, and pessimism can only switch the sender from trusting to caution.

(4) $V_T = T(1)$ is linear in ω_o^S with coefficient $u(\hat{y}_S(1)) = u(\hat{r}(1))$, and V_C is a maximum of functions linear in ω_o^S, hence convex, so G is concave in ω_o^S and the trust set is an interval. As $\omega_o^S \to \infty$ the cautious marginal payoff $-\omega_o^S u'(1-s) + \omega_e^S (m-1) u'(q) + \omega_d^S (m+1)\big(1-(m+1)s\big)$ tends to $-\infty$ at every $s > 0$ (using $u' \geq u'(1) > 0$), so $s_C \to 0$ and, by the envelope theorem, $\partial G/\partial \omega_o^S \to u(\hat{r}(1)) - u(1)$, whose sign is that of $\hat{r}(1) - 1$; the conclusions follow as in the proof of Corollary II.2. □

Proof of Proposition II.16

(1) On the trusting region the receiver's believed first-order condition is $\hat{\omega}_o^R u'(ms - \hat{r}) = \hat{\omega}_d^R (s - \hat{r})$, so $\hat{c}(s) \equiv s - \hat{r}(s) = (\hat{\omega}_o^R/\hat{\omega}_d^R)\, u'(ms - \hat{r}(s))$. Since $\hat{r}_s < m$ (Proposition II.14(4)), $ms - \hat{r}(s)$ is strictly increasing in s, and u' is nonincreasing, so $\hat{c}$ is nonincreasing, strictly where $u'' < 0$. Then $\hat{y}_S(s) = 1 - s + \hat{r}(s) = 1 - \hat{c}(s)$, and the marginal repayment $\hat{r}_s = \frac{m\lambda + \hat{\omega}_d^R}{\lambda + \hat{\omega}_d^R}$ equals one if and only if $\lambda = 0$, i.e. wherever $u'' = 0$.

(2) Since $u \geq 0$ on $[0, m]$ and $\hat{y}_S(s), q(s) \in [0, m]$, the first two terms of T lie in $[0, B]$ with $B = (\omega_o^S + \omega_e^S)\, u(m)$, which does not depend on ω_d^S. The believed gap $\hat{d}(s) = 1 - (m+1)s + 2\hat{r}(s)$ is continuous, equals $d_0 \equiv 1 - (m+1)\hat{s}_I > 0$ at the cutoff, and equals $2\hat{r}(1) - m$ at $s = 1$, which is negative for $m \geq 2$ because $\hat{r}(1) < 1$ (Proposition II.14(4)): $2\hat{r}(1) - m < 2 - m \leq 0$. So $\hat{d}$ vanishes at some $s_0 \in (\hat{s}_I, 1)$, where $T(s_0) \geq 0$. On the cautious region $\hat{d}(s) = 1 - (m+1)s \geq d_0 > 0$, so $\max_{[0,\hat{s}_I]} T \leq B - \frac{\omega_d^S}{2} d_0^2 < 0 \leq T(s_0)$ once $\omega_d^S > 2B/d_0^2$: every optimal send is then trusting. Moreover any optimal send s_+ has $T(s_+) \geq T(s_0) \geq 0$, hence $|\hat{d}(s_+)| \leq \sqrt{2B/\omega_d^S}$; since $\hat{d}(\hat{s}_I) = d_0 > 0$ and $\hat{d}(1) < 0$ are fixed, for ω_d^S large enough this forces s_+ away from both $\hat{s}_I$ and 1: the optimal send is interior, with $\hat{r}(s_+) > 0$. Under correct beliefs the realized return is $r^*(s_+) \in (0, ms_+)$ by Proposition II.14(4). □